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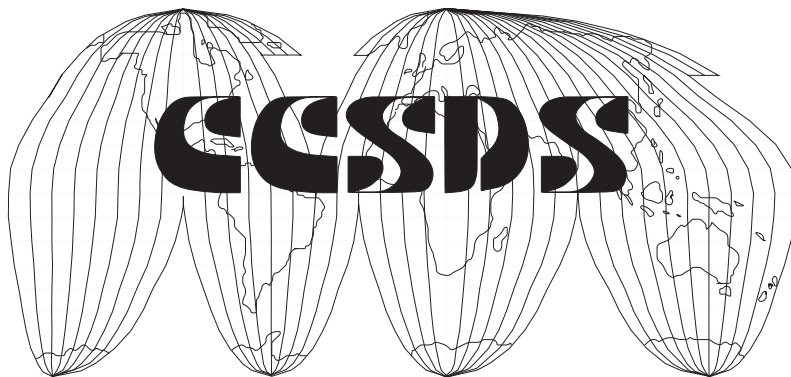
**DRAFT RECOMMENDATION FOR SPACE
DATA SYSTEM STANDARDS**

SPACE LINK EXTENSION— RETURN CHANNEL FRAMES SERVICE SPECIFICATION

CCSDS 911.2-R-1.7

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FOREWORD

(WHEN THIS RECOMMENDATION IS FINALIZED, IT WILL CONTAIN THE FOLLOWING FOREWORD:)

This document is a technical **Recommendation** for use in developing ground systems for space missions and has been prepared by the **Consultative Committee for Space Data Systems** (CCSDS). The Space Link Extension Return Channel Frames Service described herein is intended for missions that are cross-supported between Agencies of the CCSDS.

This **Recommendation** specifies a data service that extends certain of the space-to-ground communications services previously defined by CCSDS (references [2], [3], and [4]) within the framework established by the CCSDS Space Link Extension Reference Model (reference [1]). It allows implementing organizations within each Agency to proceed with the development of compatible, derived Standards for the ground systems that are within their cognizance. Derived Agency Standards may implement only a subset of the optional features allowed by the **Recommendation** and may incorporate features not addressed by the **Recommendation**.

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Questions relating to the contents or status of this document should be addressed to the CCSDS Secretariat at the address indicated on page i.

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1 INTRODUCTION

1.1 PURPOSE OF THIS RECOMMENDATION

The purpose of this Recommendation is to define the Space Link Extension (SLE) Return Channel Frames (RCF) service in conformance with the SLE Reference Model (reference [1]). The RCF service is an SLE transfer service that delivers to a mission user all telemetry frames from one master channel or one virtual channel.

NOTE – The first issue of reference [1] defines the Return Master Channel Frames (RMCF) service and the Return Virtual Channel Frames (RVCF) service as two distinct services. Subsequent study has indicated that it is preferable to define one service that provides the functionality of both. The RCF service defined here does just that. It is anticipated that the next issue of reference [1] will take the same approach, deleting the RMCF and RVCF services and replacing them with the RCF service.

1.2 SCOPE

This Recommendation defines, in an abstract manner, the RCF service in terms of:

- a) the operations necessary to provide the service;
- b) the parameter data associated with each operation;
- c) the behaviors that result from the invocation of each operation; and
- d) the relationship between, and the valid sequence of, the operations and resulting behaviors.

It does not specify:

- a) individual implementations or products;
- b) the implementation of entities or interfaces within real systems;
- c) the methods or technologies required to acquire telemetry frames from signals received from a spacecraft;
- d) the methods or technologies required to provide a suitable environment for communications; or
- e) the management activities required to schedule, configure, and control the RCF service.

1.3 APPLICABILITY

1.3.1 APPLICABILITY OF THIS RECOMMENDATION

This Recommendation provides a basis for the development of real systems that implement the RCF service. Implementation of the RCF service in a real system additionally requires the availability of a communications service to convey invocations and returns of RCF

service operations between RCF service users and providers. This Recommendation requires that such a communications service must ensure that invocations and returns of operations are transferred:

- a) in sequence;
- b) completely and with integrity;
- c) without duplication;
- d) with flow control; and
- e) with notification to the application layer in the event that communications between the RCF service user and the RCF service provider are disrupted, possibly resulting in a loss of data.

It is the specific intent of this Recommendation to define the RCF service in a manner which is independent of any particular communications services, protocols, or technologies.

1.3.2 LIMITS OF APPLICABILITY

This Recommendation specifies the RCF service that may be provided by an SLE System for inter-Agency cross support. It is neither a specification of, nor a design for, real systems that may be implemented for the control and monitoring of existing or future missions.

1.4 RATIONALE

The goal of this Recommendation is to increase the level of interoperability between the tracking stations or ground data handling systems of various Agencies and the consumers of spacecraft telemetry.

1.5 DOCUMENT STRUCTURE

1.5.1 ORGANIZATION

This document is organized as follows:

- a) Section 1 presents the purpose, scope, applicability, and rationale of this Recommendation and lists the definitions, conventions, and references used throughout the Recommendation;
- b) Section 2 provides an overview of the RCF service including a functional description, the service management context, and protocol considerations;
- c) Section 3 specifies the operations of the RCF service;
- d) Section 4 specifies the dynamic behavior of the RCF service in terms of the state transitions of the RCF service provider;

- e) Annex A provides a formal specification of RCF service data types using Abstract Syntax Notation One (ASN.1);
- f) Annex B lists all terms used in this Recommendation and identifies where they are defined;
- g) Annex C lists all acronyms used within this document;
- h) Annex D provides a conformance matrix that defines what capabilities must be provided for an implementation to be considered compliant with this Recommendation.
- i) Annex E provides a list of informative references.

1.5.2 SLE SERVICES DOCUMENTATION TREE

This Recommendation is based on the cross support model defined in the SLE Reference Model (reference [1]). It expands upon the concept of an SLE transfer service as an interaction between an SLE mission user entity (MUE) and an SLE transfer service provider for the purpose of providing the RCF transfer service.

This Recommendation is part of a suite of documents specifying the SLE services. The SLE services constitute one of the three types of Cross Support Services:

- a) Part 1: SLE Services;
- b) Part 2: Ground Domain Services
- c) Part 3: Ground Communications Services

The basic organization of the SLE services documentation is shown in figure 1-1. The various documents are described in the following paragraphs.

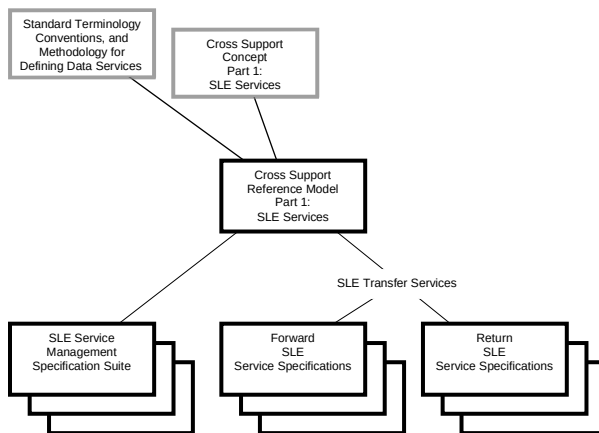


Figure 1-1: SLE Services Documentation

- a) *Standard Terminology, Conventions and Methodology for Defining Data Services* (reference [E2]); a Report identifying selected international standards relevant for the definition of data services.
- b) *Cross Support Concept—Part 1: Space Link Extension Services* (reference [E4]); a Report introducing the concepts of cross support and the SLE services.
- c) *Cross Support Reference Model—Part 1: Space Link Extension Services* (reference [1]); a Recommendation that defines the framework and terminology for the specification of SLE services.
- d) *SLE Return Service Specifications*; a set of Recommendations that will provide specification of all return link SLE services. This Recommendation is one of the specifications in that set.
- e) *SLE Forward Service Specifications*; a set of Recommendations that will provide specification of all forward link SLE services.
- f) *SLE Service Management Specifications*; a set of Recommendations that establish the basis of SLE service management. Reference [E3] is the primary Recommendation in that set.

1.6 DEFINITIONS, NOMENCLATURE, AND CONVENTIONS

1.6.1 DEFINITIONS

1.6.1.1 Definitions from OSI Basic Reference Model

This Recommendation makes use of a number of terms defined in reference [6]. The use of those terms in this Recommendation shall be understood in a generic sense; i.e., in the sense that those terms are generally applicable to technologies that provide for the exchange of information between real systems. Those terms are:

- a) abstract syntax;
- b) application entity;
- c) application layer;
- d) application process;
- e) concatenation;
- f) flow control;
- g) Open System Interconnection (OSI);
- h) real system;
- i) service access point (SAP);
- j) transfer syntax.

1.6.1.2 Definitions from Abstract Syntax Notation One

This Recommendation makes use of the following terms defined in reference [7]:

- a) Abstract Syntax Notation One (ASN.1);
- b) object identifier;
- c) (data) type;
- d) (data) value.

NOTE – In Annex A of this Recommendation, ASN.1 is used for specifying the abstract syntax of RCF service operations. The use of ASN.1 as a descriptive language is intended to support the specification of the abstract RCF service; it is not intended to constrain implementations. In particular, there is no requirement for implementations to employ ASN.1 encoding rules. ASN.1 is simply a convenient tool for formally describing the abstract syntax of RCF service operations.

1.6.1.3 Definitions from Abstract Service Definition Conventions

This Recommendation makes use of the following terms defined in reference [8]:

- a) initiator;
- b) responder.

1.6.1.4 Definitions from Directory Service Models

This Recommendation makes use of the following terms defined in reference [9]:

- a) distinguished name (DN);
- b) relative distinguished name (RDN).

1.6.1.5 Definitions from Telemetry Channel Coding

This Recommendation makes use of the following terms defined in reference [2]:

- a) attached sync marker;
- b) codeblock;
- c) convolutional code;
- d) pseudo-randomization;
- e) Reed-Solomon check symbols;

- f) Reed-Solomon code.

1.6.1.6 Definitions from Packet Telemetry

This Recommendation makes use of the following terms defined in reference [3]:

- a) frame error control field;
- b) transfer frame.

1.6.1.7 Definitions from AOS Architecture Specification

This Recommendation makes use of the following terms defined in reference [4]:

- a) coded virtual channel data unit (CVCDU);
- b) cyclic redundancy code (CRC);
- c) VCDU error control field;
- d) virtual channel data unit (VCDU).

1.6.1.8 Definitions from SLE Reference Model

This Recommendation makes use of the following terms defined in reference [1]:

- a) abstract binding;
- b) abstract object;
- c) abstract port;
- d) abstract service;
- e) invoker;
- f) Mission Data Operation System (MDOS);
- g) Mission User Entity (MUE);
- h) offline delivery mode;
- i) online delivery mode;
- j) operation;
- k) performer;
- l) physical channel;
- m) return data;

- n) Return Channel Frames channel (RCF channel);
- o) Return Channel Frames service (RCF service);
- p) service provider (provider);
- q) service user (user);
- r) SLE Complex;
- s) SLE Complex Management;
- t) SLE data channel;
- u) SLE functional group (SLE-FG);
- v) SLE protocol data unit (SLE-PDU);
- w) SLE service data unit (SLE-SDU);
- x) SLE service package;
- y) SLE System;
- z) SLE transfer service instance;
- aa) SLE transfer service production;
- bb) SLE transfer service provision;
- cc) SLE Utilization Management;
- dd) space link;
- ee) space link data channel;
- ff) space link data unit (SL-DU);
- gg) space link session.

1.6.1.9 Additional Definitions

1.6.1.9.1 Association

An association is a cooperative relationship between an SLE service-providing application entity and an SLE service-using application entity. An association is formed by the exchange of SLE protocol control information through the use of an underlying communications service.

1.6.1.9.2 Communications Service

A communications service is a capability that enables an SLE service-providing application entity and an SLE service-using application entity to exchange information.

NOTE – If an SLE service user and an SLE service provider are implemented using different communications services, then interoperability between them is possible only by means of a suitable gateway. Adherence to this Recommendation ensures, at least in principle, that it is possible to construct such a gateway.

1.6.1.9.3 Confirmed Operation

A confirmed operation is an operation that requires the performer to return a report of its outcome to the invoker.

1.6.1.9.4 Error Control Field

The error control field of a telemetry frame is the frame error control field (reference [3]) if the frame is a transfer frame or the VCDU error control field (reference [4]) if the frame is a VCDU or CVCDU.

1.6.1.9.5 Frame Version Number

The frame version number is either the transfer frame version number (reference [3]) or the version number in the VCDU primary header (reference [4]).

NOTE – The definitions of frame version number given in references [3] and [4] are equivalent. If a CCSDS-compatible telemetry frame is known to contain no errors, the frame version number enables one to distinguish between a transfer frame and a VCDU.

1.6.1.9.6 Invocation

The invocation of an operation is the making of a request by an object (the invoker) to another object (the performer) to carry out the operation.

1.6.1.9.7 Master Channel

The sequence of all telemetry frames with the same frame version number and the same spacecraft identifier on the same physical channel constitutes a master channel.

NOTE – Depending on the frame version number, the definition of spacecraft identifier is as given in either reference [3] or reference [4].

1.6.1.9.8 Parameter

A parameter of an operation is data that may accompany the operation's invocation or return.

NOTE – The term parameter is also used to refer to mission-dependent configuration information used in the production or provision of the service.

1.6.1.9.9 Performance

The performance of an operation is the carrying out of the operation by an object (the performer).

1.6.1.9.10 Port Identifier

A port identifier identifies a logical channel in a communications service. A port identifier contains all the addressing information necessary to make a connection possible.

NOTE – Port identifiers are technology-specific. For example, a port identifier might be the combination of an Internet Protocol (IP) network address and a Transport Control Protocol (TCP) port number or the combination of an OSI network address and an associated set of service access points (SAPs).

1.6.1.9.11 Return

The return of an operation is a report, from the performer to the invoker, of the outcome of the performance of the operation.

1.6.1.9.12 Service Instance Provision Period

A service instance provision period is the time during which a service instance (i.e., the capability to transfer one or more SLE data channels of a given type) is scheduled to be provided.

1.6.1.9.13 Spacecraft Identifier

The spacecraft identifier of a frame is as defined in reference [3] if the frame is a transfer frame or as defined in reference [4] if the frame is a VCDU or CVCDU.

1.6.1.9.14 Telemetry Frame (Frame)

A telemetry frame is a transfer frame (as defined in reference [3]) or a VCDU or CVCDU (as defined in reference [4]).

1.6.1.9.15 Unconfirmed Operation

An unconfirmed operation is an operation that does not require a report of its outcome to be returned to the invoker by the performer.

1.6.1.9.16 Virtual Channel

The sequence of all telemetry frames with the same frame version number, the same spacecraft identifier, and the same virtual channel identifier on the same physical channel constitutes a virtual channel.

NOTE – Depending on the frame version number, the definitions of spacecraft identifier and virtual channel identifier are as given in either reference [3] or reference [4].

1.6.1.9.17 Virtual Channel Identifier

The virtual channel identifier of a frame is as defined in reference [3] if the frame is a transfer frame or as defined in reference [4] if the frame is a VCDU or CVCDU.

1.6.2 NOMENCLATURE

The following conventions apply throughout this Recommendation:

- a) the words ‘shall’ and ‘must’ imply a binding and verifiable specification;
- b) the word ‘should’ implies an optional, but desirable, specification;
- c) the word ‘may’ implies an optional specification;
- d) the words ‘is’, ‘are’, and ‘will’ imply statements of fact.

1.6.3 CONVENTIONS

1.6.3.1 Specification of Operations

1.6.3.1.1 General

Section 3 of this Recommendation specifies the operations that comprise the RCF service. The specification of each operation is divided into subsections as follows:

1.6.3.1.2 Purpose Subsection

The Purpose subsection provides a brief description of the purpose of the operation. Additionally, it indicates whether the operation may be invoked by the user, provider, or both; whether the operation is confirmed or unconfirmed; and whether there are any constraints on when the operation may be invoked.

1.6.3.1.3 Invocation, Return, and Parameters Subsection

The Invocation, Return, and Parameters subsection describes the parameters associated with each operation, including their semantics. A table accompanying the description of each operation lists all parameters associated with the operation and, for both the invocation and return, whether the parameter is always present, always absent, or conditionally present.

For parameters that are conditionally present, the parameter description specifies the conditions for the presence or absence of the parameter. The condition is generally based on the value of another parameter in the same invocation or return; for example, the parameter `diagnostic` is present in a return if and only if the value of the parameter `result` is 'negative result'. For a conditional parameter in a return, the condition may be based on the value of a parameter in the corresponding invocation.

In the table, the following convention is used to indicate whether a parameter is always present, always absent, or conditionally present:

M	Always present
C	Conditionally present
Blank	Always absent

NOTE – Even though a parameter may be characterized as always present, its description may specify that its value is permitted to be 'null' or 'unused' or the like.

1.6.3.1.4 Effects Subsection

The Effects subsection describes the effects an operation has on the invoker, the performer, the association between them, or any combination thereof. The details of how those effects occur or the mechanisms used are outside the scope of this Recommendation.

1.6.3.2 Typographic Conventions

1.6.3.2.1 Operation Names

Names of RCF service operations always appear in uppercase and begin with the characters 'RCF-' (e.g., RCF-TRANSFER-DATA).

1.6.3.2.2 Parameter Names

Names of parameters of RCF service operations generally appear in lowercase and are always typeset in a fixed-width font (e.g., `responder-port-identifier`).

1.6.3.2.3 Value Names

The values of many parameters discussed in this Recommendation are represented in the text by names. Those names are always shown in quotation marks (e.g., ‘no such service instance’). The actual value associated with the name is constrained by the type of the parameter taking on that value. Parameter types are specified in Annex A of this Recommendation.

NOTES

- 1 The name of a value does not imply anything about its type. For example, the value ‘no such service instance’ has the appearance of a character string but might be assigned to a parameter whose type is ‘integer’.
- 2 Contrary to traditional publication style, sentence punctuation, such as a comma or period, follows the closing quotation mark of a value name rather than preceding it.

1.6.3.2.4 State Names

This Recommendation specifies the states of RCF service providers. States may be referred to by number (e.g., state 2) or by name. State names are always shown in quotation marks (e.g., ‘active’).

1.6.3.2.5 Data Type Definitions

Data type definitions for the RCF service are presented in Annex A in the form of a set of ASN.1 modules. Regardless of the conventions used elsewhere in this Recommendation, the text of the ASN.1 modules is typeset entirely in a fixed-width font.

1.6.3.3 Other Conventions

This Recommendation uses the conventions specified in reference [1].

1.7 REFERENCES

The following documents contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All documents are subject to revision, and users of this Recommendation are encouraged to investigate the possibility of applying the most recent editions of the documents indicated below. The CCSDS Secretariat maintains a register of currently valid CCSDS Recommendations.

- [1] *Cross Support Reference Model—Part 1: Space Link Extension Services.*
Recommendation for Space Data System Standards, CCSDS 910.4-B-1. Blue Book.
Issue 1. Washington, D.C.: CCSDS, May 1996.

- [2] *Telemetry Channel Coding*. Recommendation for Space Data System Standards, CCSDS 101.0-B-3. Blue Book. Issue 3. Washington, D.C.: CCSDS, May 1992.
- [3] *Packet Telemetry*. Recommendation for Space Data System Standards, CCSDS 102.0-B-4. Blue Book. Issue 4. Washington, D.C.: CCSDS, November 1995.
- [4] *Advanced Orbiting Systems, Networks and Data Links: Architectural Specification*. Recommendation for Space Data Systems Standards, CCSDS 701.0-B-2. Blue Book. Issue 2. Washington, D.C.: CCSDS, November 1992.
- [5] *Time Code Formats*. Recommendation for Space Data System Standards, CCSDS 301.0-B-2. Blue Book. Issue 2. Washington, D.C.: CCSDS, April 1990.
- [6] *Information technology—Open Systems Interconnection—Basic Reference Model: The Basic Model*. International Standard, ISO/IEC 7498-1:1994. Geneva: ISO, 1994.
- [7] *Information technology — Open Systems Interconnection — Specification of Abstract Syntax Notation One (ASN.1)*. International Standard, ISO/IEC 8824:1990. 2nd ed. Geneva: ISO, 1990.
- [8] *Information technology—Text Communication—Message-Oriented Text Interchange Systems (MOTIS) —Part 3: Abstract Service Definition Conventions*. International Standard, ISO/IEC 10021-3:1990. Geneva: ISO, 1990.
- [9] *Information technology—Open Systems Interconnection—The Directory—Part 2: Models*. International Standard, ISO/IEC 9595-2:1993. Geneva: ISO, 1993.

2 DESCRIPTION OF THE RETURN CHANNEL FRAMES SERVICE

2.1 OVERVIEW

The RCF service enables the user of the service to obtain all telemetry frames from one master channel or one virtual channel. A master channel consists of all telemetry frames with the same frame version number and the same spacecraft identifier on the same physical channel. A virtual channel consists of all telemetry frames with the same frame version number, the same spacecraft identifier, and the same virtual channel identifier on the same physical channel. A telemetry frame is a transfer frame, a virtual channel data unit (VCDU), or a coded virtual channel data unit (CVCDU). A space link physical channel carries one stream of telemetry frames separated by attached sync markers. A physical channel may be comprised of one or more master channels, each of which may be comprised of one or more virtual channels. A complete specification of these concepts is provided in references [2], [3], and [4].

For delivery to the user, each frame acquired from the space link is encapsulated in an SLE service data unit (SLE-SDU) that also carries annotation, e.g., the Earth Receive Time (ERT) of the frame. In general, the RCF service delivers frames to the user in the order in which they were received from the space link.

The operations defined in section 3 of this Recommendation enable an RCF service user to interact with an RCF service provider to:

- a) establish an association between the user and the provider;
- b) receive annotated telemetry frames from one master channel or from one virtual channel from the provider;
- c) obtain notifications and reports regarding the status, configuration and performance of the service;
- d) temporarily suspend and later re-start the delivery of telemetry frames;
- e) change the values of certain parameters that affect the behavior of the service; and
- f) release an association.

The provision of RCF service for one master channel or one virtual channel for access by one service user constitutes one instance of service. The provision of RCF service for one master channel or one virtual channel to multiple service users and the provision of RCF service for multiple master channels or multiple virtual channels to one or more service users are permitted but are specified to constitute multiple service instances.

2.2 SPACE LINK EXTENSION REFERENCE MODEL

2.2.1 INTRODUCTION

The RCF service is specified within the framework defined by the SLE Reference Model (reference [1]). The SLE Reference Model is based, in part, on the Abstract Service Definition Conventions (ASDC) defined in reference [8]. The following paragraphs summarize selected concepts from the SLE Reference Model.

2.2.2 ABSTRACT OBJECT

An abstract object is a functional entity that interacts with other abstract objects. Objects are of different types, which determine their function and behavior. Objects are characterized by their interfaces, which are called abstract ports, and the operations that are made available through those interfaces.

2.2.3 ABSTRACT SERVICE

An abstract service is the capability provided by a set of operations that an abstract object exposes at one or more of its abstract ports.

NOTE – The concept of an abstract service is to be distinguished from the concept of an (N)-service as defined in the OSI Basic Reference Model (reference [6]). The definition of (N)-service is in terms of the capability provided by one layer in the OSI architecture to the layer above it. The definition of abstract service is in terms of the capability provided by one abstract object to another abstract object. In a cross support scenario where one Agency is providing an SLE service to another Agency, the object that provides the service typically is associated with one Agency, and the object that uses the service typically is associated with the other Agency.

2.2.4 SERVICE USER/PROVIDER

An object that offers a service to another by means of one or more of its ports is called a service provider (provider). The other object is called a service user (user). An object may be a provider of some services and a user of others.

The terms user and provider are used to distinguish the roles of two interacting objects. In this Recommendation, when two objects are involved in provision of a service, the object closer to the space link is considered to be the provider of the service, and the object further from the space link is considered to be the user.

2.2.5 ABSTRACT BINDING

When two abstract ports of the same type have an association established between them such that an abstract service can be provided, the two ports are said to be bound. The act of establishing such an association is called abstract binding.

The terms initiator and responder are used to describe the interaction between two objects with respect to abstract binding. One object (the initiator) invokes a bind operation which is accepted (or rejected) by another object (the responder).

2.2.6 OPERATION

An operation is a procedure or task that one object (the invoker) can request of another (the performer) through a port pair bound within the terms of an agreement.

The terms invoker and performer are used to describe the interaction between two objects as the operations that constitute the service occur. One object invokes an operation that is performed by the other. For most services, each object invokes some operations and performs others.

2.3 SERVICE MANAGEMENT

SLE service management determines the number, type, and schedule of RCF service instances to be provided, the resources required to enable those service instances, and the initial configuration of all service instances and their supporting resources. SLE service management is the subject of separate CCSDS Recommendations, and the CCSDS Secretariat should be contacted for information regarding the status of those Recommendations.

The SLE Reference Model (reference [1]) distinguishes between service provision and service production. Certain configuration parameters are associated with provision of RCF service while others are associated with production. Changes to RCF provision configuration parameters (e.g., quality of service) affect only a single service instance; the values of such parameters are initialized by service management when the service instance is created but may be modified subsequently by the user through RCF service operations specified in this Recommendation. Changes to RCF production configuration parameters (e.g., bit rate, frame length, coding type) potentially affect multiple service instances or potentially impact SLE Complex resources; consequently, those parameters are modified only through service management.

In general, RCF service may be user-initiated (i.e., the user invokes the bind operation) or provider-initiated (i.e., the provider invokes the bind operation). However, any particular service instance shall support either user initiation or provider initiation but not both. Which form of initiation applies to a particular service instance is set by service management.

The SLE Reference Model defines two data delivery modes: online deliver mode and offline deliver mode. Online delivery mode indicates that the provision of service is generally coincident in time with the space link session, whereas offline delivery mode indicates that the telemetry frames acquired during a space link session are provided to the user some time after the end of the space link session. Within the online delivery mode, the RCF service recognizes two variations: timely online deliver mode and complete online delivery mode. Both the timely and complete online delivery modes assume the use of a reliable communications service. They differ in that the timely mode allows for the controlled discarding of telemetry frames at the application layer if it is not possible to deliver those frames within a certain amount of time after they are acquired from the space link (e.g., due to communications service backlog). While the RCF service is defined for the complete online delivery mode, the timely online delivery mode, or the offline delivery mode, any particular instance of RCF service shall support only one of those modes. The delivery mode applicable to a particular service instance is set by service management.

2.4 ARCHITECTURE MODEL—FUNCTIONAL VIEW

2.4.1 RETURN SPACE LINK PROCESSING FUNCTIONAL GROUP

The Return Frame Processing Functional Group (RFP-FG, shown in figure 2-1) is the SLE functional group (SLE-FG) that produces the RCF service. As shown in the figure, the RFP-FG provides other services in addition to the RCF service. This Recommendation addresses only the RCF service; the other services are to be defined in companion Recommendations.

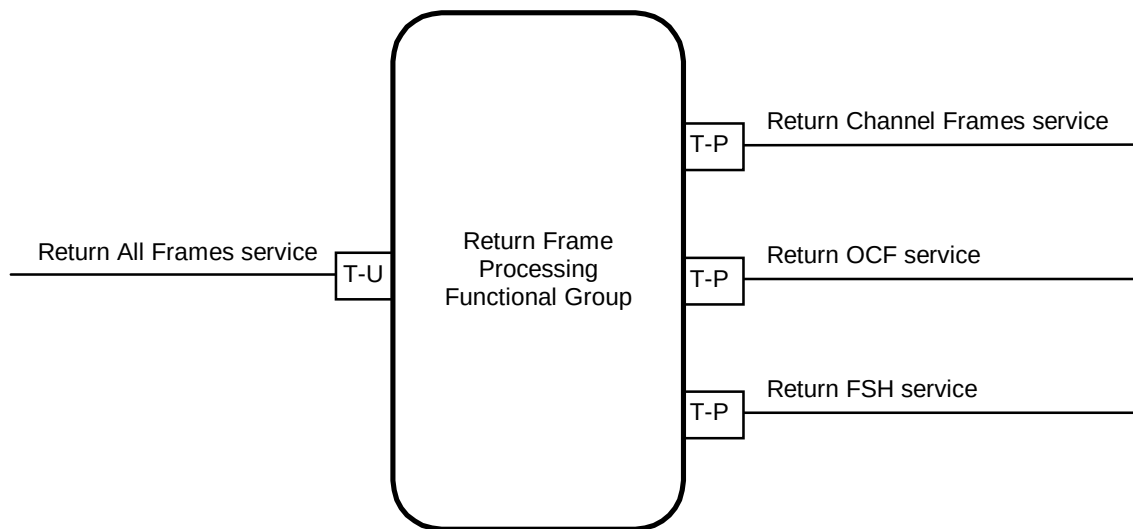


Figure 2-1: Return Frame Processing SLE-FG

As described in reference [1], the RFP-FG consumes consumes Return All Frames (RAF) service and provides RCF service. By means of the consumed RAF service, the RFP-FG receives an RAF channel. The RAF channel consists of a stream of SLE-SDUs that encapsulate the telemetry frames acquired from one space link physical channel. From this input, the RFP-FG produces one or more RCF channels. Each RCF channel consists of a

stream of SLE-SDUs. Most of these SLE-SDUs encapsulate telemetry frames from the master channel or virtual channel specified by the user of the RCF service; such SLE-SDUs also carry annotation information associated with the frame (e.g., the ERT of the frame). Other SLE-SDUs in an RCF channel carry notification of the occurrence of certain events that may pertain to the RCF service (e.g., loss of frame synchronization on the physical channel associated with this instance of RCF service). An RCF channel produced by the RFP-FG is delivered to a user by means of the RCF service. More specifically, the RFP-FG performs the following functions with respect to RCF service:

- a) consumes one RAF channel;
- b) demultiplexes the RAF channel into its component master channels, demultiplexes the master channels into their component virtual channels, annotates each frame in each channel to form RCF SLE-SDUs, and injects the resulting RCF SLE-SDU into RCF channels;
- c) optionally, stores (and subsequently retrieves) sufficient data to reconstruct the RCF channels for delivery through one or more offline RCF service instances;
- d) makes RCF channels available to online and offline RCF service instances to effect the provision of RCF service.

The telemetry frames delivered by the RCF service are encapsulated in SLE-SDUs. RCF SLE-SDUs that encapsulate telemetry frames are annotated with information that pertains to that specific frame. The annotation consists of:

- a) the ERT of the frame;
- b) an identifier that indicates the antenna used to acquire the frame;
- c) a parameter that characterizes the data link continuity of this frame with respect to the preceding frame on the space link;
- d) an optional octet string that may be used to provide additional, non-standard annotations that are mutually agreed to by the SLE Complex providing the service and the Mission Data Operations System (MDOS) associated with the user of the service.

NOTES

- 1 The RCF service delivers only telemetry frames that are error-free. The determination that a frames is error-free is based on the frame quality annotation provided by the RAF service: a frame is considered error-free if it was annotated by the RAF service with a frame quality of 'good'. The RAF service annotates a frame as 'good' if the frame contains only valid codewords of the Reed-Solomon code or—if the frame is not Reed-Solomon encoded—if the frame error control field decodes successfully.

- 2 If the RCF service delivers frames from a master channel or virtual channel that was Reed-Solomon encoded on the space link, the Reed-Solomon check symbols are removed from the frame before it is delivered.
- 3 As defined by this Recommendation, telemetry frames do not include the attached sync marker but are delimited by them.

2.4.2 RCF SERVICE PRODUCTION AND PROVISION

One instance of RCF service production (or, one RFP-FG instance) may be associated with multiple RCF service instances. RCF production is concerned with the production of RCF channels independent of any particular instance of service. In contrast, RCF service provision is concerned with delivering an RCF channel to an RCF service user. Service provision addresses such matters as when service is provided (e.g., service start and stop times), how service is provided (e.g., user-initiated or provider-initiated), and quality of service (e.g., whether the delivery mode is timely online, complete online, or offline).

The SLE-SDUs generated by RCF service production are delivered to the service user by means of the RCF service operations defined in section 3, which also provide additional functionality to facilitate the provision of RCF service. In turn, the RCF service operations are realized as SLE protocol data units (SLE-PDUs) that are exchanged between the RCF service provider and the RCF service user by means of an underlying communications service. The relationship between SLE-SDUs and SLE-PDUs is illustrated in figure 2-2.

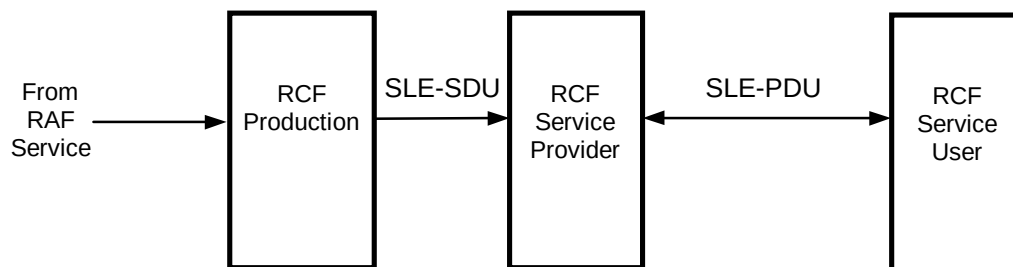


Figure 2-2: RCF Service Production and Provision

For the online delivery mode, production and provision of the RCF service by the provider occur, at least in part, concurrently with the space link session. For the offline delivery mode, service production and provision are detached, with service provision occurring some time after the end of the space link session. In the offline case, data acquired during the space link session are stored for later delivery by an offline service instance.

2.5 ARCHITECTURE MODEL—CROSS SUPPORT VIEW

The management and control of the production and provision of SLE transfer services is described in general terms in reference [1]. Figure 2-3 shows an example operational scenario and the related binding of RCF transfer service ports (defined in this

Recommendation) and the SLE management port (outside the scope of this Recommendation, but see reference [E3]). This scenario shows an SLE Complex with one RFP-FG instance providing two instances of RCF service to a Mission Data Operations System.

NOTE – Although not shown in this scenario, many other combinations are possible. For example, it is also possible to have several RFP-FG instances, each consuming a different RAF channel and each providing one or more instances of service. It is also possible for the Return Space Link Processing Functional Group (RSLP-FG) and the RFP-FG to be located in different SLE Complexes.

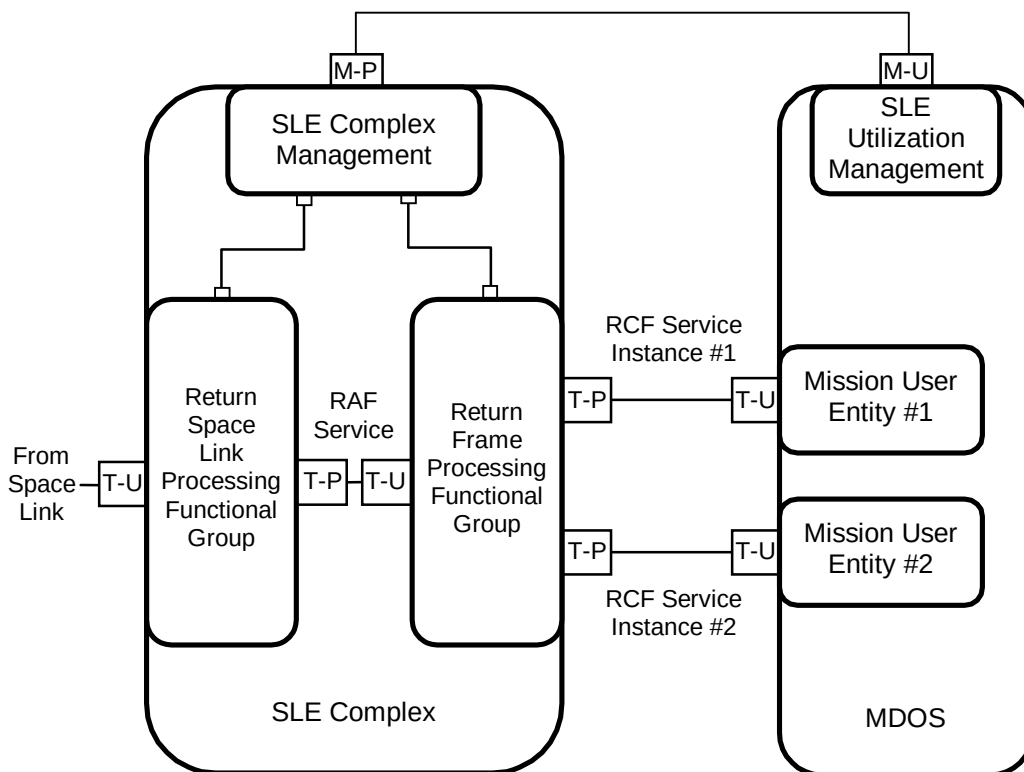


Figure 2-3: Example of the Management and Provision of RCF Service

2.6 FUNCTIONAL DESCRIPTION

2.6.1 GENERAL

This subsection describes the RCF service with respect to scheduling, configuration, underlying services, provider states, and protocol considerations.

2.6.2 SCHEDULING AND CONFIGURATION

SLE Utilization Management negotiates with SLE Complex Management to establish mutually agreed upon SLE service packages. Among other things, SLE service packages specify what service instances are to be provided, when those services are to be provided, and what resources are needed to enable those services.

Service packages also specify the initial values of mission-dependent parameters required for service production and provision. RCF service provision parameters include such things as the scheduled start and stop times of the RCF service instance and the delivery mode.

Service production is guaranteed to occur only as needed to support service packages that have been scheduled and mutually agreed upon by SLE Complex Management and SLE Utilization Management. Service provision occurs only within the bounds of the agreed upon schedule of service instances and only during those periods when there is an association between the service provider and the service user.

2.6.3 UNDERLYING SERVICES

The RCF service is based on the functionality provided by the SLE RAF service (reference [1]) or an equivalent capability. The RAF service (or its equivalent) may be provided by the same SLE Complex that provides the RCF service or by a different SLE Complex. Additionally, provision of RCF service depends on service management for scheduling, resources, and configuration, and on the availability of a suitable communications service to enable the exchange of information between the RCF service user and provider.

2.6.4 PROTOCOL DESCRIPTION

2.6.4.1 RCF Operations

The operations that constitute the RCF service are listed in table 2-1. Section 3 of this Recommendation contains the detailed specification of these operations.

Table 2-1: RCF Operations

Operation	Invoked By	Purpose
RCF-BIND	User or provider	To establish an association with the peer
RCF-UNBIND	User or provider	To release an association previously established by an RCF-BIND operation
RCF-START	User	To request that the SLE service provider start the delivery of telemetry frames
RCF-STOP	User	To request that the SLE service provider stop the delivery of telemetry frames

Operation	Invoked By	Purpose
RCF-TRANSFER-DATA	Provider	To transfer a telemetry frame to the SLE service user
RCF-SYNC-NOTIFY	Provider	To notify the user of an event affecting production or provision of the RCF service
RCF-SCHEDULE-STATUS-REPORT	User	To request that the provider send a status report immediately or periodically or to stop such reporting
RCF-STATUS-REPORT	Provider	To send a status report to the user
RCF-GET-PARAMETER	User	To ascertain the value of certain SLE service parameters
RCF-PEER-ABORT	User or Provider	To notify the peer that the local SLE application detected an error that requires the association to be terminated

2.6.4.2 States of the Service Provider

Once an RCF service instance is created, the RCF service provider is in one of three states, as follows:

- a) State 1 ('unbound'): In state 1, all resources required to enable the provision of the RCF service have been allocated, and all objects required to provide the service have been instantiated. However, no association yet exists between the user and the provider (i.e., the RCF transfer service provider port is not bound).
- b) State 2 ('ready'): In state 2, an association has been established between the user and the provider, and they may interact by means of the operations described in section 3 of this Recommendation. However, the delivery of telemetry frames (by means of the RCF-TRANSFER-DATA operation) is not permitted. The user may enable the delivery of telemetry frames by means of the appropriate service operation (RCF-START); that, in turn, will cause the provider to transition to state 3 ('active') and enable frame delivery.
- c) State 3 ('active'): State 3 is identical to state 2 ('ready') except that all telemetry frames are delivered to the user as they become available. The service continues in this state until the user invokes the RCF-STOP operation to suspend frame delivery and transition back to state 2 (e.g., in response to an 'end of data' notification from the service provider signaling that the space link session has ended and all available frames have been delivered).

A simplified RCF service provider state transition diagram is shown in figure 2-4. A detailed state transition matrix is provided in section 4.

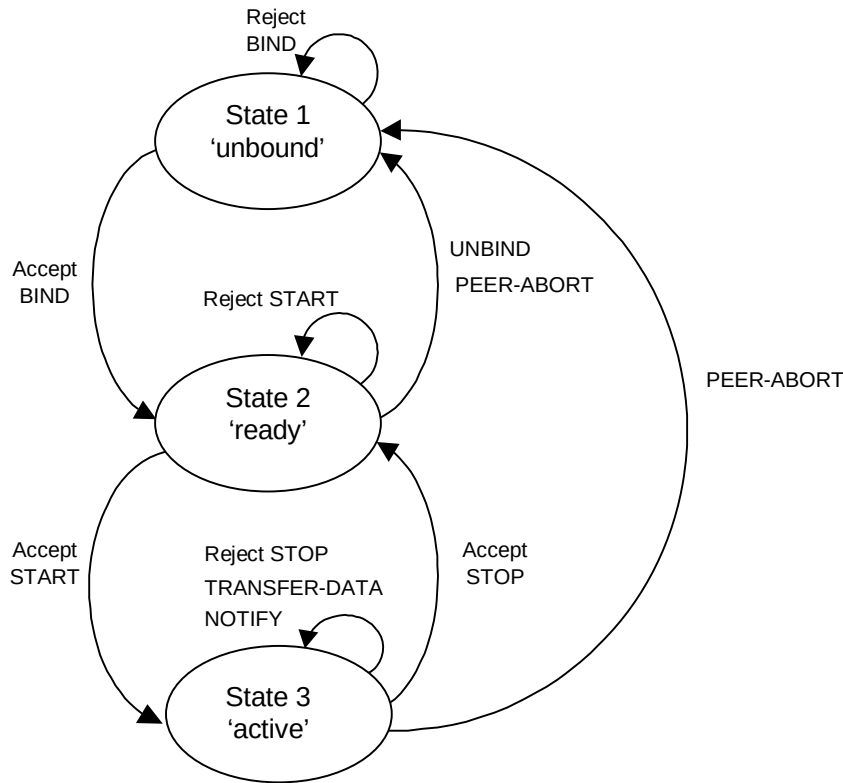


Figure 2-4: Simplified RCF Service Provider State Transition Diagram

2.6.4.3 Termination of Association

An association is released normally when an RCF-UNBIND operation is invoked by the initiator of the association and performed by the responder. An association may be aborted by either the user or the provider by means of the RCF-PEER-ABORT operation. An association also may be aborted due to certain failures of the communications service; such failures are signaled to the local application by the 'protocol abort' event described in section 4.

2.6.4.4 Effects of Termination

When an association is released or aborted, no further operations shall be invoked by the user or the provider. As a consequence, the delivery of frames stops immediately. The user and provider may re-establish an association via a new RCF-BIND operation if that is consistent with the schedule for the provision of service. However, status information from the prior association is not preserved and is not available to the new association except that:

- a) if the delivery mode is complete online, the contents of the online frame buffer (see 3.1.8) shall be persistent;

- b) if the delivery mode is offline, the contents of the offline frame buffer (see 3.1.8) shall be persistent;
- c) statistics reported by means of the RCF-STATUS-REPORT operation (see 3.9) shall be accumulated for the entire service instance provision period.

2.6.4.5 Technology Aspects

This Recommendation defines the RCF service. Provision of the RCF service in a real system also requires a specification of how the RCF service defined here is mapped to a communications service such that invocations and returns of RCF service operations can be conveyed between the user and the provider. In order not to restrict the applicability of this Recommendation to a specific communications technology, as few assumptions as possible have been made about the characteristics of the underlying communications service (see 1.3.1).

The service interface between the user and the provider is specified in this Recommendation in terms of the operations that the service provides. However, implementation of the service in real systems requires a mapping of the service operations to protocol data units that can be exchanged by means of the communications service. Within the scope of this Recommendation, RCF service operations are mapped to SLE-PDUs in accordance with the buffering mechanisms described in 3.1.8; e.g., multiple RCF-TRANSFER-DATA operations may be mapped to a single SLE-PDU. From the point of view of the provider or user application, the interaction is in terms of operations; but from the point of view of the application entities that implement the protocol, what is exchanged are SLE-PDUs that may represent multiple RCF service operations. The mapping of service operations to SLE-PDUs is illustrated in figure 2-5.

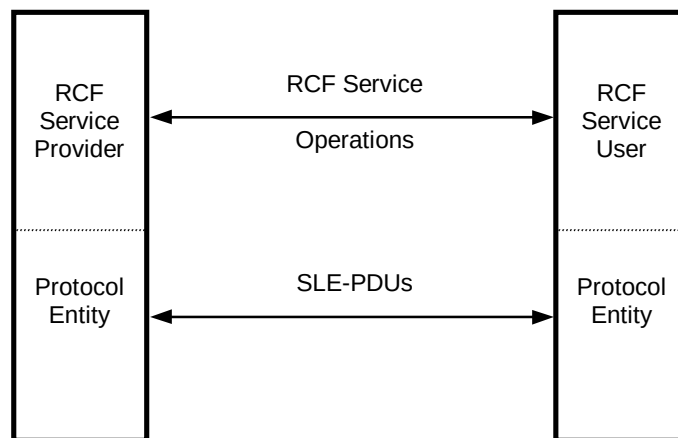


Figure 2-5: Mapping of RCF Service Operations to SLE-PDUs

The situation is complicated further when the underlying communications service is considered. Ideally, interoperability is achieved by the user and the provider agreeing to use the same communications technology. However, such an agreement may not always be possible. In the case where the peer applications have been built on top of different communication technologies, it is still possible to achieve interoperability, but only by means of a gateway. Figure 2-6 shows a user and a provider that use different communications technologies and interoperate through a gateway. The RCF SLE-PDUs are translated on one side into forms appropriate for a particular communications technology. These forms are then translated by the gateway into forms appropriate for another communications technology. Finally, those forms are translated back into RCF SLE-PDUs.

NOTE – In order to accomplish interoperability, even in the case of implementations using the same communications service, there is still the need to agree on how to map the SLE-PDUs into forms appropriate for that communications service.

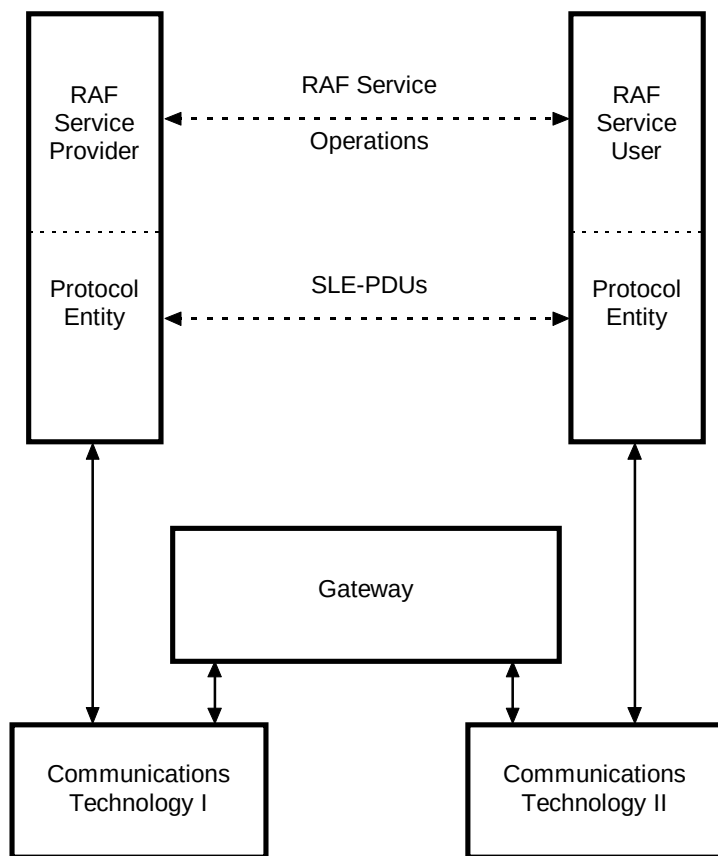


Figure 2-6: RCF Service Provision Via a Gateway

Because the operations of the RCF service are relatively simple, once an association is in place between the service user and the service provider, the technology specific elements involved in the exchange of SLE-PDUs are relatively minor. However, the way an

association is established (i.e., the binding) tends to be specific to the communications technology in use. Nonetheless, the RCF-BIND and RCF-UNBIND operations as specified in this document are intended to be ‘technology neutral’. This neutrality is achieved by specifying that the values of some parameters of the bind operation are logical names. The conversion of logical names to technology specific parameters is a local matter. The conversion methodology may rely on simple techniques such as look-up tables or may use more elaborate mechanisms such as naming or directory services. The gateway must be able to reconcile differences between technology specific parameters on either side of this gateway. The key parameter in this regard is the **responder-port-identifier**, which is discussed in the following paragraphs.

NOTE – Conceptually, it is useful to view the gateway as being part of either the SLE Complex or the MDOS. In terms of management, the parameters relevant for the binding are communicated in the form of the communications technology employed by the system which does not include the gateway. It is the responsibility of the gateway-hosting system to convert the parameters submitted by management to the ‘local’ form.

The value of the **responder-port-identifier** parameter is the logical name of the location of the peer SLE application to which an initiator is requesting to bind. After conversion to the technology specific form, the **responder-port-identifier** serves a dual purpose:

- a) The **responder-port-identifier** parameter identifies a ‘physical’ address that must be known so that the communications provider can route SLE-PDUs as required. If a gateway is involved, the conversion of the logical name at the initiating side must result in the ‘physical’ address of the gateway. As part of the conversion by the gateway, the gateway’s address is replaced by the address of the responding SLE system.

NOTE – The term ‘physical’ address is used here in a generic sense and not necessarily limited to the identification of a physical port. Depending on the technology, it may also encompass elements such as a TCP port number, OSI selectors, etc.

- b) The **responder-port-identifier** also identifies the responder application’s location. A gateway may have to adapt this part to the technology specific presentation expected at the receiving end, but logically this part of the port identifier is not modified by a gateway.

Binding is accomplished only when an association between the peer applications is established. A communications gateway implementation must ensure this. In particular, a gateway shall only return a report of the outcome of the RCF-BIND operation to the initiator when it has received such a return from the responder. It is not sufficient to establish an association between the initiator and the gateway.

A gateway must also ensure the consistency of the state of the association as perceived by both the user and the provider. In particular, in case the association with the RCF application on one side of the gateway is aborted (e.g., due to a communications service fault), the gateway must ensure that the association with the RCF application on the other side of the gateway is also aborted.

2.6.4.6 Buffering

Buffering mechanisms used by the RCF protocol are described in the following paragraphs. They are formally specified by the requirements in 3.1.8 and the state transition matrix in section 4.

2.6.4.6.1 Transfer Buffer

As described in 2.6.4.5, RCF operations (specified in section 3) are mapped to SLE-PDUs (specified in Annex A) that are conveyed to the peer SLE entity by means of the underlying communications service. In general, there is a one-to-one mapping between SLE-PDUs and the invocations or returns of RCF operations. However, that is not always the case. In particular, the RCF protocol provides that multiple RCF-TRANSFER-DATA and RCF-SYNC-NOTIFY operations may be mapped to a single SLE-PDU (viz., the SLE-PDU named *RcfTransferBuffer* in Annex A). In terms of the RCF service, the release of the *RcfTransferBuffer* to the communications service provider is equivalent to the near-simultaneous invocation of multiple RCF-TRANSFER-DATA and RCF-SYNC-NOTIFY operations. These operations are invoked in the order in which the original annotated telemetry frames and synchronous notifications occur. However, the RCF service provider concatenates them in a buffer, the content of which forms a single *RcfTransferBuffer* SLE-PDU. This SLE-PDU is the service data unit passed to the communications provider.

The primary rationale for this approach is as follows: when the data rate on the space link exceeds the available communications bandwidth or when the ground communication link is congested or unavailable for a period of time, use of the complete online delivery mode may lead to the accumulation of a large backlog of undelivered data, resulting in the delivery of data past the point of usefulness. The timely online delivery mode is an alternative that limits the size of the backlog that is allowed to accumulate by discarding data that cannot be delivered within a certain time. Furthermore, when data is discarded, it is discarded “in chunks”, i.e., as a sufficiently large block of contiguous frames rather than as random frames here and there. In general, this approach maximizes the usefulness of the data that is delivered.

This result is achieved as follows: as the RCF service provider acquires telemetry frames and as events that must be synchronously notified occur, that information is stored in a buffer named the transfer buffer. The size of this buffer is set by service management to achieve the appropriate level of “chunking” of data; this size corresponds to the maximum-sized *RcfTransferBuffer* SLE-PDU that will be passed to the communications service provider. (This size must also be compatible with the size of the service data units that can be handled by the underlying communications service.) When data is inserted into the transfer buffer, if

the transfer buffer was previously empty, a timer, named the release timer, is started. The release timer counts down from an initial time value, named the latency limit, that is set by service management. If the transfer buffer becomes full or if the release timer expires, the entire transfer buffer, in the form of one RcfTransferBuffer SLE-PDU, is passed to the communications service provider as one service data unit. If new data needs to be inserted into the transfer buffer, but the transfer buffer is full and cannot be passed to the communications service provider due to congestion of the communications service, then the entire transfer buffer is discarded as one unit. When this happens, the new data is inserted into the newly-emptied transfer buffer, along with a synchronous notification the some data was intentionally discarded due to timeliness considerations.

Strictly speaking, the transfer buffer is only required in the case of timely online delivery mode. However, the transfer buffer mechanism has a secondary benefit: it allows for “tuning” of the size of the service data units that are normally passed to the communications service provider. In some cases, this may contribute to enabling the communications service to operate more efficiently. Thus, the transfer buffer is used in all delivery modes. However, in the case of complete online delivery mode and offline delivery mode, data in the transfer buffer is never discarded. Rather, the contents of the transfer buffer are held until they can be passed to the communications service provider. Any backlog that may accumulate is handled by means of the online frame buffer or offline frame buffer (see 2.6.4.6.2).

2.6.4.6.2 Online Frame Buffer and Offline Frame Buffer

In the case of the timely online delivery mode, the only buffering that occurs is that provided by the transfer buffer. For complete online delivery mode and for offline delivery mode, additional buffering is need due to the nature of the service. Since the complete online delivery mode is supposed to deliver all data even in the case of extended communications service outages or congestion and since the offline delivery mode is supposed to deliver all data even several days after the space link session, more extensive buffering is required for these modes. This buffering is provided by means of the online frame buffer or the offline frame buffer, respectively. These buffers are relatively large and intended to hold all data (i.e., RCF-TRANSFER-DATA and RCF-SYNC-NOTIFY) for significant periods of time. In the case of complete online delivery mode, the online frame buffer is intended to overcome limitations of the communications service: bandwidth limitations, outages, and congestion. In the case of offline delivery mode, the offline frame buffer is intended to enable data to be delivered hours or days after the completion of the space link session. The exact size of these buffers is set by service management. It is normally expected that the online frame buffer is sufficiently large to hold all data that might be accumulated during one space link session and that the offline frame buffer is sufficiently large to hold all data that might be accumulated during several space link sessions.

NOTE – Synchronous notifications are generally not provided in offline mode, so the data associated with such notifications does not need to be stored in the offline frame buffer.

Figure 2–7 illustrates the differences between timely online delivery mode, complete online deliver mode, and offline delivery mode with respect to the buffers that are used.

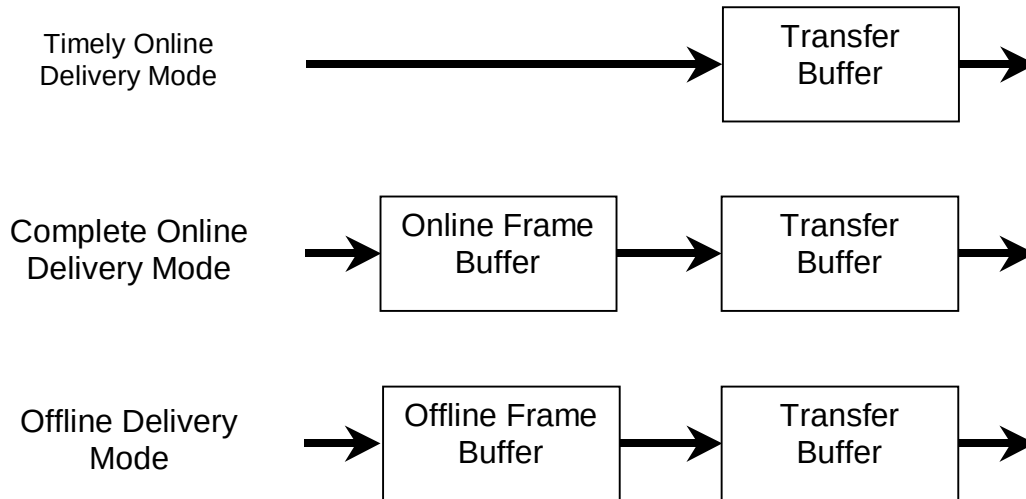


Figure 2-7. Buffers and Delivery Modes

2.7 OPERATIONAL SCENARIO

The following paragraphs illustrate a typical sequence of operations in the online delivery mode between the user and the provider of the RCF service.

- a) Prior to the actual provision of service, start and stop times for both the space link session and the associated RCF service instance are mutually agreed upon by SLE Complex Management and SLE Utilization Management. Configuration and other information needed to enable the service are also agreed upon.
- b) Some time before the scheduled start time of the RCF service instance, the service instance is created by SLE Complex Management. Initially, the service provider is in state 1 ('unbound'). At the scheduled start time of the space link session, the SLE Complex acquires the signal from the spacecraft and initiates the production of RCF service. Typically (but not necessarily) the start time of the service instance will precede by a small margin the start time of the space link session to allow the user to bind to the service before the start of the space link session.
- c) The user invokes the RCF-BIND operation to establish an association.
- d) The provider transitions from state 1 to state 2 ('ready') and returns a report of the successful completion of the bind operation to the user.
- e) The user may now check parameters that control service provision by means of the RCF-GET-PARAMETER operation.
- f) If the user is interested in obtaining periodic status reports, the RCF-SCHEDULE-STATUS-REPORT operation may be invoked to configure status reporting.

- g) The user invokes the RCF-START operation to enable data flow.
- h) The provider transitions from state 2 to state 3 ('active') and confirms the RCF-START operation to the user. As telemetry frames that meet the delivery criteria established by the user become available, they are delivered to the user by means of RCF-TRANSFER-DATA operations. In addition, notifications may be delivered by means of RCF-SYNC-NOTIFY operations, and requested status reports are delivered by means of RCF-STATUS-REPORT operations.
- i) When all available frames have been delivered, the provider will invoke the RCF-SYNC-NOTIFY operation to provide an 'end of data' notification. The 'end of data' notification may be triggered, for example, because the space link session ended and all frames have been delivered or because the user specified a value for the `stop-time` parameter when invoking the RCF-START operation.
- j) By invoking the RCF-STOP operation the user will cause the provider to transition to state 2 ('ready'). The user may then ask for another set of frames by invoking another RCF-START command or may terminate the association by invoking RCF-UNBIND.

3 RCF SERVICE OPERATIONS

NOTE – This section (3) specifies the processing of valid SLE-PDUs (i.e., those that are recognized as the invocation or return of an RCF service operation). Subsection 3.1 specifies behaviors that are generally applicable to all operations. Subsections 3.2 through 3.11 specify individual operations. Handling of invalid SLE-PDUs is specified in section 4.

3.1 GENERAL CONSIDERATIONS

3.1.1 RESULT OF OPERATIONS

3.1.1.1 All confirmed operations shall report on the outcome of the operation in a return.

3.1.1.2 All returns shall include a `result` parameter that indicates whether the outcome of the operation was successful ('positive result') or unsuccessful ('negative result').

3.1.1.3 In the event of a 'negative result', the return shall also include a `diagnostic` parameter, the value of which is descriptive of the reason for the 'negative result'.

NOTE – Possible values of the `diagnostic` parameter are listed in the description of each operation.

3.1.1.4 A `diagnostic` parameter value of 'other' shall be returned only if no other value in the list adequately describes the reason for the 'negative result'.

3.1.2 PARAMETER TYPES

The types of all parameters shall conform to the abstract syntax specified in Annex A.

NOTE – Some parameter types in Annex A are chosen such that possible future extensions of the range of allowed values of a parameter will not cause a type mismatch. For example, parameters that logically are of the 'enumerated' type may be specified as being of the 'named integer' type.

3.1.3 PARAMETER CHECKING

3.1.3.1 Validity checks shall be performed on the values of parameters associated with an operation.

NOTE – Rules governing the validity of parameter values are included in the specification of individual operations. General reasons for regarding a parameter value as invalid are specified in the following paragraphs.

3.1.3.2 A parameter value shall be treated as invalid if its value is outside the range or not in the set of values currently permitted by service management for the given parameter.

NOTE – A conforming implementation shall be capable of supporting the full range or set as specified in Annex A.

3.1.3.3 A parameter value shall be treated as invalid if its value is in conflict with the value of another value in the same invocation.

NOTE – For example, the value of the `start-time` parameter in an invocation of RCF-START is invalid if it is later than the value of the `stop-time` parameter.

3.1.3.4 If a parameter value is not valid, the operation shall not be performed, and, for confirmed operations, a report of ‘negative result’ shall be returned to the invoker.

3.1.3.5 Unless explicitly specified otherwise, parameter checks should be performed in the order in which diagnostics are listed in the descriptions of the operations, and the `diagnostic` parameter should be set to the value defined for the first problem found.

3.1.3.6 In the case that an implementation does not adhere to the sequence of checks as specified by the sequence of diagnostics values, such implementation shall specify the sequence in which the checks are actually performed.

3.1.4 ACCESS CONTROL

3.1.4.1 For each instance of RCF service, service management shall specify the authorized initiator and responder.

3.1.4.2 When processing an RCF-BIND invocation, the responder shall verify that the initiator, as indicated by the `initiator-identifier` parameter of the invocation, is the authorized initiator for this service instance; if not, the responder shall not perform the RCF-BIND and shall return a ‘negative result’ to the initiator.

3.1.4.3 When processing the return from RCF-BIND, the initiator shall verify that the responder, as indicated by the `responder-identifier` parameter of the return, is the authorized responder for this service instance; if not, the initiator shall abnormally terminate the association by invoking RCF-PEER-ABORT.

3.1.5 AUTHENTICATION

NOTE – Requirements for security depend on the application and the SLE system environment (e.g., whether closed or public networks are used or if access is only from physically restricted areas). In many environments, security may be provided by the communications service, transparently to the SLE application. This Recommendation does not preclude the use of security features that are provided by the communications service or the local environment, nor does it assume the availability of such features.

3.1.5.1 The RCF service shall provide the following options with respect to the level of authentication of invocations and returns of operations:

- a) 'all': all RCF invocations and returns, except the invocation of RCF-PEER-ABORT, shall be authenticated;
- b) 'bind': only the RCF-BIND invocation and return shall be authenticated;
- c) 'none': no RCF invocations or returns shall be authenticated.

3.1.5.2 SLE Complex Management and SLE Utilization Management shall agree on the level of authentication to be required for an association between a service user and a service provider and shall configure both entities accordingly.

3.1.5.3 SLE Complex Management and SLE Utilization Management shall agree on the algorithm used to generate and check credentials parameters and shall make this algorithm known to the service user and service provider together with associated parameters such as passwords or keys as necessary for the selected algorithm.

NOTES

- 1 The specification of the algorithms themselves is outside the scope of this Recommendation.
- 2 The `initiator-identifier` and `responder-identifier` parameters of the RCF-BIND operation identify the user and provider and therefore the applicable authentication level and algorithm necessary to generate and check credentials.

3.1.5.4 For operations for which authentication is required by the terms of the agreement between SLE Complex Management and SLE Utilization Management:

- a) invocations shall include an `invoker-credentials` parameter to permit the performer to authenticate the invocation;
- b) returns shall include a `performer-credentials` parameter to permit the invoker to authenticate the return.

3.1.5.5 For operations for which authentication is not required, the `invoker-credentials` or `performer-credentials` parameter should be set to the value 'unused' to signify that the invocation or return does not carry credentials.

3.1.6 BLOCKING AND NON-BLOCKING OPERATIONS

3.1.6.1 To support applications that may need to invoke several operations concurrently, the parameter `invoke-ID` is specified for all confirmed operations except RCF-BIND and RCF-UNBIND.

NOTES

- 1 The **invoke-ID** parameter allows the invoker to correlate a particular return to the invocation that prompted it.
- 2 Confirmed operations that include the **invoke-ID** parameter are non-blocking operations; those that do not are blocking operations. Unconfirmed operations are always non-blocking.

3.1.6.2 After invoking a blocking operation, the invoker shall not invoke another operation for the same service instance until the return from the blocking operation is received; except that, if the return is not received in a timely manner, the invoker may invoke RCF-PEER-ABORT to terminate the association.

3.1.6.3 After invoking a non-blocking operation, the invoker may invoke another operation without waiting for the return from the first invocation.

3.1.6.4 The value of the **invoke-ID** parameter shall be an invoker-supplied arbitrary integer value that shall be returned, unchanged, by the performer.

3.1.6.5 An error condition shall exist if an invocation includes an **invoke-ID** whose value is the same as that of another invocation that is awaiting confirmation within the context of the same service instance.

3.1.6.6 To ensure that the RCF service behaves in a predictable manner, the effects of operations shall be as though the operations were performed in the order that their invocations were received by the performer.

3.1.6.7 The invoker may choose not to exploit the non-blocking capability and always wait for the return from a non-blocking operation before invoking another operation.

NOTE – An invoker wishing to operate in blocking mode (i.e., to invoke a new operation only when the return from the previous operation has been received) may use a constant value for the **invoke-ID** parameter. As long as a return is still outstanding, the performer will reject any further invocations.

3.1.6.8 Compliance with this Recommendation does not require the performer to process invocations concurrently; however, the performer must accept invocations from a non-blocking invoker and buffer and serialize them by local means not visible externally.

3.1.7 TIME

Coordinated Universal Time (UTC) shall be used for all parameters containing a time value.

3.1.8 DELIVERY MODES

3.1.8.1 Timely Online Delivery Mode

3.1.8.1.1 For timely online delivery mode, the RCF service provider shall store successfully acquired frames and associated annotations in a buffer in the form of records of the `RcfTransferDataInvocation` type (see Annex A). This buffer shall be called the transfer buffer. The annotation shall be as defined in 3.6.

3.1.8.1.2 Upon the occurrence of any of several events that cause a change to or disruption of the data being provided to the service instance, the RCF service provider shall store a synchronous notification record of the event in the transfer buffer in the form of an `RcfSyncNotifyInvocation` record. The notification record shall be stored following the last annotated frame acquired before the event and before the first annotated frame acquired following the event. The events that must be stored are as defined in 3.7.

3.1.8.1.3 If the transfer buffer was empty before the service provider inserted an `RcfTransferDataInvocation` or `RcfSyncNotifyInvocation` record into the transfer buffer, the service provider shall start a timer for the transfer buffer. This timer shall be named the release timer.

3.1.8.1.4 The duration from the time that the release timer is started until it expires is given by the parameter `latency-limit`, the value of which shall be set by service management.

3.1.8.1.5 The size of the transfer buffer for a given instance of RCF service shall be set by service management.

3.1.8.1.6 The contents of the buffer shall be passed to the communications service (in the form of one `RcfTransferData` SLE-PDU) and the transfer buffer shall be cleared as soon as one of the following conditions is met:

- a) the buffer is full (i.e., no more `RcfTransferDataInvocation` or `RcfSyncNotifyInvocation` records can be accommodated in the transfer buffer); or
- b) the release timer expires.

3.1.8.1.7 The `RcfTransferBuffer` SLE-PDU shall contain the records in the same sequence as they were inserted into the transfer buffer.

3.1.8.1.8 If the underlying communications service generates back pressure due to congestion (e.g., it does not accept the `RcfTransferBuffer` SLE-PDU or would block the RCF service provider from continuing), the RCF service provider shall discard this `RcfTransferBuffer` SLE-PDU and clear the transfer buffer. It shall then insert a 'data discarded due to excessive backlog' `RcfSyncNotifyInvocation` record into the transfer buffer and restart the release timer.

3.1.8.1.9 In the timely online delivery mode, the transfer buffer shall be cleared:

- a) whenever the user stops data flow by invoking the RCF-STOP operation; or
- b) whenever the association is aborted.

3.1.8.1.10 Only RCF-TRANSFER-DATA and RCF-SYNC-NOTIFY invocations shall be buffered through the transfer buffer. The invocations or returns of all other operations shall be asynchronous (in the sense that they are not required to be invoked or returned in sequence with respect to the delivery of telemetry frames). Therefore, they shall be invoked as soon as possible without regard to the contents of the transfer buffer.

3.1.8.2 Complete Online Delivery Mode

3.1.8.2.1 For complete online delivery mode, the RCF service provider shall store successfully acquired frames and associated annotations in a large circular buffer in the form of records of the RcfTransferDataInvocation type (see Annex A). This buffer shall be called the online frame buffer. The annotation shall be as defined in 3.6.

NOTE – Complete online delivery mode attempts to deliver all acquired frames, in order, with minimum delay consistent with the available ground communications bandwidth. Complete online delivery requires that the online frame buffer be sufficiently large to deal with communications service delays, outages, and bandwidth limitations.

3.1.8.2.2 Upon the occurrence of any of several events that cause a change to or disruption of the data being provided to the service instance, the RCF service provider shall store a synchronous notification record of the event in the online frame buffer in the form of an RcfSyncNotifyInvocation record. The notification record shall be stored following the last annotated frame acquired before the event and before the first annotated frame acquired following the event. The events that must be stored are as defined in 3.7.

3.1.8.2.3 For complete online delivery mode, the size of the online frame buffer and the transfer buffer shall be determined by arrangement between SLE Complex Management and SLE Utilization Management and shall be set by service management.

3.1.8.2.4 The RCF service provider shall start to fill the online frame buffer as soon as both the service instance provision period and the space link session have started and frames are being acquired from the space link.

3.1.8.2.5 The RCF service provider shall extract frames and synchronous notifications from the online frame buffer and insert them in the same sequence into the transfer buffer in the form of RcfTransferDataInvocation and RcfSyncNotifyInvocation records until either the transfer buffer is full or the notification inserted flags an ‘end of data’ event.

3.1.8.2.6 As soon as the transfer buffer is full or an ‘end of data’ RcfSyncNotifyInvocation record has been inserted into the transfer buffer, the service provider shall build from the transfer buffer contents an RcfTransferBuffer SLE-PDU and shall attempt to pass this SLE-PDU to the communications service until it is accepted.

3.1.8.2.7 When the communications service provider has accepted the RcfTransferBuffer SLE-PDU, the RCF service provider shall clear the transfer buffer, discard frames and notifications that were contained in the transfer buffer from the online frame buffer, and resume extracting frames and synchronous notifications from the online frame buffer as described above.

3.1.8.2.8 The RCF service provider shall continue to extract frames and notifications from the online frame buffer to the transfer buffer and to pass the contents of the transfer buffer to the communications service as long as the service instance remains in state 3 ('active').

3.1.8.2.9 The RCF service provider shall continue to stored incoming annotated frames and notification records into the online frame buffer until the end of the service instance provision period, regardless of the state of the service instance and regardless of whether an association with the service user is established.

3.1.8.2.10 In the case that the user invokes the RCF-STOP operation or the association becomes unbound, the user may, after re-binding if necessary, invoke a new RCF-START operation, with a start time in the past, to effect delivery of data buffered in the online frame buffer. Any frames with an ERT older than the start time specified in the RCF-START operation and any notifications falling into the same interval shall be removed from the online frame buffer.

3.1.8.2.11 If the online frame buffer becomes full (e.g., because an extended communications outage prevents it from being drained), the provider shall discard annotated frame and notification records from the online frame buffer in oldest-first order. The number of frames to be discarded in such event is set by service management. The RCF service provider shall also clear the transfer buffer and then insert an RcfSyncNotifyInvocation record indicating a 'data discarded' event. Extraction of frames and notifications from the online frame buffer shall then resume as before.

NOTE – For the complete online delivery mode, it is intended that the size of the online frame buffer be selected such that overflow of the buffer is a very rare event.

3.1.8.2.12 If the service user, in the RCF-START invocation, requests a start time earlier than any frame still held in the online frame buffer, the provider shall deliver the earliest data available.

3.1.8.2.13 At the end of the scheduled service instance provision period, the contents of the online frame buffer shall be discarded.

3.1.8.2.14 Only RCF-TRANSFER-DATA and RCF-SYNC-NOTIFY invocations shall be buffered through the online frame buffer and the transfer buffer. The invocations or returns of all other operations shall be asynchronous (in the sense that they are not required to be invoked or returned in sequence with respect to the delivery of telemetry frames). Therefore, they shall be invoked as soon as possible without regard to the contents of the online frame buffer or the transfer buffer.

3.1.8.3 Offline Delivery Mode

3.1.8.3.1 Frames that are to be provided through an offline delivery mode service instance shall be acquired during the space link session and held in a large circular buffer, called the offline frame buffer.

3.1.8.3.2 Offline service shall not be provided until sometime after the end of the space link session.

3.1.8.3.3 When an RCF-START operation is invoked in the context of an offline delivery mode RCF service instance, the RCF service provider shall extract annotated frames from the offline frame buffer and insert them into the transfer buffer in the form of RcfTransferDataInvocation records. Such extraction shall begin with the frame in the offline frame buffer with the earliest ERT that is later than the start time designated in the RCF-START invocation.

3.1.8.3.4 Extraction of annotated frames from the offline frame buffer and their insertion into the transfer buffer shall continue, with subsequent frames being extracted from the offline frame buffer in the same order in which they were originally received from the space link physical channel. Such extraction shall continue until:

- a) a frame is retrieved with an ERT which is later than the stop time in the RCF-START invocation (in which case an 'end of data' notification shall be generated and inserted into the transfer buffer);
- b) the user invokes the RCF-STOP operation; or
- c) the association is aborted.

3.1.8.3.5 Whenever the transfer buffer is full or an 'end of data' notification is inserted into the transfer buffer, the RCF service provider shall build an RcfTransferBuffer SLE-PDU from the contents of the transfer buffer and shall attempt to pass this SLE-PDU to the communications service until it is accepted. The RcfTransferBuffer SLE-PDU shall contain the RcfTransferDataInvocation records in the same sequence as the corresponding frames were retrieved from the offline frame buffer.

3.1.8.3.6 Once the RcfTransferBuffer SLE-PDU has been accepted by the communications service, the RCF service provider shall clear the transfer buffer and resume extracting frames from the offline frame buffer.

3.1.8.3.7 For complete online delivery mode, the size of the online frame buffer and the transfer buffer shall be determined by arrangement between SLE Complex Management and SLE Utilization Management and shall be set by service management.

3.1.8.3.8 In the case that the user invokes the RCF-STOP operation or the association becomes unbound, the user may, after re-binding if necessary, invoke a new RCF-START operation, specifying a new ERT interval for which frames shall be delivered from the offline frame buffer. The start and stop times of such an RCF-START invocation may be earlier,

later, or the same as the start and stop times of any previous RCF-START invocation, provided that they are valid start and stop times as specified in 3.4.

3.1.8.3.9 If the user, in the RCF-START invocation, requests a start time earlier than any frame still held in the offline frame buffer, the provider shall deliver frames beginning with the earliest data available.

3.1.8.3.10 Except for ‘end of data’ notifications as described above, synchronous notifications shall not be provided in the offline delivery mode.

3.1.8.3.11 Only RCF-TRANSFER-DATA invocations and the ‘end of data’ RCF-SYNC-NOTIFY invocation shall be buffered through the offline frame buffer and the transfer buffer. The invocations or returns of all other operations shall be asynchronous (in the sense that they are not required to be invoked or returned in sequence with respect to the delivery of telemetry frames). Therefore, they shall be invoked as soon as possible without regard to the contents of the offline frame buffer or the transfer buffer.

3.1.8.3.12 During offline service provision, the RCF service provider shall invoke successive RCF-TRANSFER-DATA operations without undue delay, subject only to limitations imposed by the underlying communications service or to any maximum data rate limitation (“metering”) that may be imposed through service management.

3.2 RCF-BIND

3.2.1 PURPOSE

3.2.1.1 The RCF-BIND operation shall be used to establish an association between the initiator and the responder.

3.2.1.2 For every instance of RCF service, service management shall establish whether that instance of service is to be user-initiated or provider-initiated:

- a) for a user-initiated service instance, only the service user is permitted to invoke RCF-BIND;
- b) for a provider-initiated service instance, only the service provider is permitted to invoke RCF-BIND.

3.2.1.3 The responder shall return a report of the outcome of the performance of the RCF-BIND operation to the initiator.

3.2.1.4 Except as provided in 3.2.1.5, the invoker of RCF-BIND shall not invoke any further operations for this service instance until the return from the responder is received.

3.2.1.5 If the return from RCF-BIND is not received after a sufficiently long time, the initiator may attempt to recover by invoking the RCF-PEER-ABORT operation followed by another RCF-BIND.

NOTE – The length of the duration that constitutes “a sufficiently long time” is determined by service management.

3.2.1.6 The RCF-BIND operation is valid only in state 1 (‘unbound’).

3.2.2 INVOCATION, RETURN, AND PARAMETERS

3.2.2.1 General

The parameters of the RCF-BIND operation shall be present in the invocation and return as specified in table 3-1.

3.2.2.2 **invoker-credentials**

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the RCF-BIND invocation (see 3.1.5).

Table 3-1: RCF-BIND Parameters

Parameter	Invocation	Return
invoker-credentials	M	
performer-credentials		M
initiator-identifier	M	
responder-identifier		M
responder-port-identifier	M	
service-type	M	
version-number	M	C
service-instance-identifier	M	
result		M
diagnostic		C

3.2.2.3 performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of RCF-BIND (see 3.1.5).

3.2.2.4 initiator-identifier

The **initiator-identifier** parameter shall identify the authority on whose behalf the initiating SLE application is initiating the association.

NOTES

- 1 The **initiator-identifier** parameter permits the responder to determine if the RCF-BIND operation is being invoked by the authorized initiator for this service instance.
- 2 Each value of the **initiator-identifier** parameter is associated with exactly one authentication level and exactly one authentication scheme.
- 3 If authentication based on credentials is used, this parameter may be redundant since the **initiator-identifier** value may be one constituent of the **invoker-credentials** parameter. However, the encoding may differ, and it may be convenient to have this parameter available in “clear text” form.

3.2.2.5 responder-identifier

The **responder-identifier** parameter shall identify the authority on whose behalf the responding SLE application is acting.

NOTES

- 1 The **responder-identifier** parameter permits the initiator to determine if the RCF-BIND return is from the authorized responder for this service instance.
- 2 Each value of the **responder-identifier** parameter is associated with exactly one authentication level and exactly one authentication scheme.
- 3 If authentication based on credentials is used, this parameter may be redundant since the **responder-identifier** value may be one constituent of the **performer-credentials** parameter. However, the encoding may differ, and it may be convenient to have this parameter available in “clear text” form.

3.2.2.6 responder-port-identifier

The **responder-port-identifier** parameter shall specify the logical name of the location of the responding SLE application with which the initiator seeks to establish an association.

NOTES

- 1 SLE complex management and SLE utilization management must have previously agreed on the set of valid logical names and their translations and on the logical name applicable for this service instance.
- 2 The translation of the logical name must provide sufficient information in an appropriate form to permit establishment of a connection with the named location using the agreed upon communications service.

3.2.2.7 service-type

The **service-type** parameter shall specify the type of service that will be provided if the bind operation succeeds.¹

¹ For the RCF-BIND operation, the **service-type** parameter is redundant, because the only valid value of **service-type** is ‘Return Channel Frames’. However, it is anticipated that future work by CCSDS may result in RCF-BIND being superseded by a generic SLE-BIND operation that is invoked with any one of several SLE service types. The RCF-BIND **service-type** parameter is provided in an attempt to facilitate such a change.

3.2.2.8 **version-number**

3.2.2.8.1 The **version-number** parameter shall identify the version number of the RCF service specification that is to govern this association if RCF-BIND succeeds.

3.2.2.8.2 **version-number** is conditionally present in the return based on the **result** parameter:

- a) if the value of **result** is 'positive result', **version-number** shall be present in the return;
- b) if the value of **result** is 'negative result', **version-number** shall not be present in the return.

3.2.2.8.3 If the value of the **result** parameter is 'positive result', the responder shall either:

- a) accept the version proposed by the initiator by putting the same version number into the return; or,
- b) if the responder supports version negotiation, propose a lower (earlier) version number by putting the lower number into the return.

3.2.2.8.4 If the responder implementation does not support the requested version and does not support a lower version (or does not support version negotiation), the responder shall reject the bind with the **diagnostic**, 'version not supported'.

3.2.2.8.5 If the responder proposes a lower version and the initiator implementation does not support version negotiation, the initiator shall unbind the association.

3.2.2.8.6 The value of the **version-number** parameter for the RCF service defined by this issue of this Recommendation shall be '1'.

NOTE – The version negotiation process as outlined above is feasible only as long as future versions of the RCF service do not modify the specification of the RCF-BIND operation.

3.2.2.9 **service-instance-identifier**

The **service-instance-identifier** parameter shall uniquely identify this service instance within the scope of the service-providing SLE Complex.

NOTE – It is recommended that the **service-instance-identifier** parameter be specified as a distinguished name (see reference [9]). In that case, the attributes of the distinguished name should include the SLE service agreement and the SLE service package applicable to this service instance. Reference [E3] specifies a suitable definition for the **service-instance-identifier** parameter.

3.2.2.10 result

The **result** parameter shall specify the result of the RCF-BIND operation and shall contain one of the following values:

- a) 'positive result'—the RCF-BIND invocation is accepted by the responder, and the association is established;
- b) 'negative result' —the RCF-BIND invocation is rejected by the responder for the reason specified in the **diagnostic** parameter, and the association is not established.

3.2.2.11 diagnostic

3.2.2.11.1 If **result** is 'negative result', **diagnostic** shall be present and shall contain one of the following values:

- a) 'access denied'—the value of the **initiator-identifier** parameter is not recognized by the responder (e.g., the value does not identify the authorized initiator of any service instance known to the responder);
- b) 'service type not supported'—the value of the **service-type** parameter of the RCF-BIND invocation does not identify a service type supported by the responder;
- c) 'version not supported'—the responder does not support the requested version, and the responder implementation does not permit version negotiation or does not support any version of the service lower than the one requested by the initiator;
- d) 'no such service instance'—the requested service instance is not defined in any agreed upon service package known to the responder;
- e) 'already bound'—the service instance is already bound via a different association.
- f) 'service instance not accessible to this initiator'—the authorized initiator for the service instance identified by the **service-instance-identifier** parameter does not match the initiator identified by the **initiator-identifier** parameter of the RCF-BIND invocation.
- g) 'inconsistent service type'—the value of the **service-type** parameter of the RCF-BIND invocation is not 'Return Channel Frames', or the value of the **service-type** parameter does not match the service type of the service instance identified by the **service-instance-identifier** parameter;
- h) 'invalid time'—the RCF-BIND operation was invoked outside the service instance provision period agreed to in the service package;
- i) 'out of service'—the responder has been taken out of service for an indefinite period by management action;

- j) 'unable to comply'—the responder is unable to accept the bind at this time because of a fault affecting the responder;
- k) 'other' —the reason for the negative result will have to be found by other means.

NOTES

- 1 In some implementations, under some circumstances, it may not be possible for the intended performer to provide a return in the event of the conditions indicated by diagnostics d), h), or i).
- 2 Implementations should consider that, under some conditions, RCF-BIND may fail with no return (e.g., if the value of the `responder-port-identifier` parameter is incorrect).

3.2.2.11.2 If `result` is 'positive result', `diagnostic` shall not be present in the return.

3.2.3 EFFECTS

3.2.3.1 If `result` is 'positive result', the RCF-BIND operation shall have the following effects:

- a) an association between the user and the provider shall be established;
- b) the provider shall transition from state 1 ('unbound') to state 2 ('ready');
- c) all service parameters shall be set to the initial values determined by service management;
- d) upon receipt of the positive return, the user may proceed to invoke other RCF service operations (e.g., to configure the service and begin data transfer).

3.2.3.2 If `result` is 'negative result', the RCF-BIND operation shall have the following effects:

- a) an association between the user and the provider shall not be established;
- b) the provider shall remain in state 1 ('unbound');
- c) upon receipt of the negative return:
 - 1) the initiator should examine the `diagnostic` parameter for the cause;
 - 2) the initiator may attempt to re-invoke the RCF-BIND.

3.3 RCF-UNBIND

3.3.1 PURPOSE

3.3.1.1 The initiator shall invoke the RCF-UNBIND operation to release an association previously established by RCF-BIND.

3.3.1.2 The responder shall return a report of the outcome of the performance of the RCF-UNBIND operation to the initiator.

3.3.1.3 Except as provided in 3.3.1.4, the initiator shall not invoke any further operations for this service instance until the return from RCF-UNBIND is received; nor shall it perform any further operations invoked by the responder; nor shall it return to the responder any further reports of the outcome of operations invoked by the responder.

NOTE – The initiator may invoke the RCF-UNBIND operation even if it did not yet receive all returns from previously invoked operations. The initiator should be aware that the responder may choose not to send any further returns as soon as it has received the RCF-UNBIND invocation. It may then happen that the RCF-UNBIND return is not received before one of the missing returns causes a ‘missing return’ timeout (see section 4).

3.3.1.4 If the return from RCF-UNBIND is not received after a sufficiently long time, the initiator should invoke the RCF-PEER-ABORT operation to abort the association.

NOTES

- 1 The length of the duration that constitutes “a sufficiently long time” is determined by service management.
- 2 Following receipt of the return from RCF-UNBIND or following the invocation of RCF-PEER-ABORT, the initiator may issue another RCF-BIND if otherwise permitted (e.g., if the end of the service instance provision period has not been reached).

3.3.1.5 The RCF-UNBIND operation is valid only in state 2 (‘ready’).

3.3.1.6 The RCF-UNBIND operation shall be invoked only by the initiator (i.e., the invoker of the RCF-BIND operation that established this association).

3.3.2 INVOCATION, RETURN, AND PARAMETERS

3.3.2.1 General

The parameters of the RCF-UNBIND operation shall be present in the invocation and return as specified in table 3-2.

Table 3-2: RCF-UNBIND Parameters

Parameter	Invocation	Return
invoker-credentials	M	
performer-credentials		M
unbind-reason	M	
result		M

3.3.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the RCF-UNBIND invocation (see 3.1.5).

3.3.2.3 performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of RCF-UNBIND (see 3.1.5).

3.3.2.4 unbind-reason

3.3.2.4.1 The **unbind-reason** parameter shall indicate the reason the RCF-UNBIND operation is being invoked.

3.3.2.4.2 If the RCF-UNBIND operation is invoked by the user, the **unbind-reason** parameter shall take one of the following values:

- a) ‘end’—the user has obtained all frames that are needed or expected and is releasing the association normally; the provider may delete the service instance and release all resources associated with it;

NOTE – If **unbind-reason** is ‘end’, any subsequent attempt to invoke RCF-BIND will fail even if the service instance provision period has not expired, since the service provider may release the resources allocated to that service instance.

- b) ‘suspend’—the user is suspending usage of this service instance for an unspecified period of time; the user may or may not re-bind to the provider to continue data transfer at some time prior to the end of the service instance provision period;
- c) ‘version not supported’—the user does not support the version of the RCF service proposed by the provider in the return from RCF-BIND; this value of **unbind-reason** shall be used only if the RCF-UNBIND is the first operation invoked following the RCF-BIND;

- d) 'other'—the reason for the release will have to be found by other means.

3.3.2.4.3 If the RCF-UNBIND operation is invoked by the provider, the **unbind-reason** parameter shall take one of the following values:

- a) 'end'—the provider has transferred all available frames to the user and is releasing the association normally; the provider shall not attempt to re-bind to the user in the context of this service instance;
- b) 'suspend'—the provider is suspending service provision for an unspecified period of time; the provider may attempt to re-bind to the user to continue data transfer at some time prior to the end of the service instance provision period;
- c) 'version not supported'—the provider does not support the version of the RCF service proposed by the user in the return from RCF-BIND; this value of **unbind-reason** shall be used only if the RCF-UNBIND is the first operation invoked following the RCF-BIND;
- d) 'other'—the reason for the release will have to be found by other means.

3.3.2.5 result

The **result** parameter shall specify the result of the RCF-UNBIND operation and shall always contain the following value:

- a) 'positive result'—the RCF-UNBIND operation has been performed by the responder, and the association is released.

NOTES

- 1 If RCF-UNBIND is invoked while the provider is not in the 'ready' state, the responder invokes RCF-PEER-ABORT; if the authentication level is 'all' and the invocation of RCF-UNBIND cannot be authenticated, the responder ignores it; but there is no situation in which the response to the invocation of RCF-UNBIND is to return 'negative result'.
- 2 The **result** parameter is returned for the RCF-UNBIND operation, even though the only permitted value is 'positive result', for consistency with other confirmed operations.
- 3 RCF-UNBIND is a confirmed operation in order to provide a definite indication to the initiator that the responder has performed the operation and the association is released.

3.3.3 EFFECTS

Performance of the RCF-UNBIND operation shall have the following effects:

- a) the association between the initiator and the responder shall be released, and the initiator and the responder shall cease to communicate with each other;
- b) the provider shall transition to state 1 ('unbound');
- c) if `unbind-reason` is 'end', the provider may delete the service instance and release its resources;
- d) If `unbind-reason` is not 'end', the initiator may attempt to re-bind at any time prior to the end of the service instance provision period.

NOTE – The performance of RCF-UNBIND for a particular service instance does not necessarily terminate RCF production (e.g., if `unbind-reason` is not 'end' and the delivery mode is complete online or if another service instance is dependent on the production).

3.4 RCF-START

3.4.1 PURPOSE

3.4.1.1 The user shall invoke the RCF-START operation to request that the provider begin the delivery of telemetry frames.

3.4.1.2 The provider shall return a report of the outcome of the performance of the RCF-START operation to the user.

3.4.1.3 Following a successful RCF-START, the provider shall deliver telemetry frames acquired from the space link to the user as quickly as those frames are available.

NOTES

1 Frames are delivered to the user by means of the RCF-TRANSFER-DATA operation (see 3.6).

2 Communications service delays may affect the rate at which available frames are delivered.

3.4.1.4 All frames delivered following the RCF-START but prior to the next RCF-STOP (see 3.5) shall be delivered in the order in which they were received from the space link.

3.4.1.5 The user may specify, as parameters of the RCF-START invocation, the ERTs of the first and last telemetry frames that are to be delivered by the provider.

3.4.1.6 The time parameters may be changed during an association by invoking an RCF-STOP followed by an RCF-START with new time parameters.

NOTE – This capability is intended primarily to support the offline delivery mode.

3.4.1.7 RCF-START is valid only in state 2 ('ready') and shall be invoked only by the user.

3.4.2 INVOCATION, RETURN, AND PARAMETERS

3.4.2.1 General

The parameters of the RCF-START operation shall be present in the invocation and return as specified in table 3-3.

3.4.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the RCF-START invocation (see 3.1.5).

Table 3-3: RCF-START Parameters

Parameter	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
start-time	M	
stop-time	M	
global-VCID	M	
result		M
diagnostic		C

3.4.2.3 performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of RCF-START (see 3.1.5).

3.4.2.4 invoke-ID

The RCF service provider shall return unchanged the user-supplied value of the **invoke-ID** parameter (see 3.1.6).

3.4.2.5 start-time

3.4.2.5.1 The value of the **start-time** parameter shall be ‘null’, or it shall be a time value that indicates that only frames with an ERT equal to or later than **start-time** shall be delivered.

3.4.2.5.2 For the online delivery mode, only frames acquired during the space link session associated with this service instance shall be delivered, regardless of the value of **start-time**.

3.4.2.5.3 For the offline delivery mode, the provider shall deliver all available frames that meet the delivery criteria regardless of the space link session in which they were acquired.

3.4.2.5.4 For the online delivery mode, if **start-time** is ‘null’, the data transfer shall begin with the next frame that is acquired from the space link.

3.4.2.5.5 For the offline delivery mode, **start-time** must not be ‘null’.

3.4.2.5.6 To be valid, **start-time** must satisfy the following criteria:

- a) for the online delivery mode, **start-time** must be equal to or later than the start time of the service instance provision period for this service instance;
- b) for the online delivery mode, **start-time** must be earlier than the end time of the service instance provision period for this service instance;
- c) **start-time** must be earlier than **stop-time** (see 3.4.2.6).

NOTE – The provider is able to deliver only frames that have been acquired from the space link. For example, in an online service instance, if **start-time** is earlier than the start time of the space link session, the first frame that is delivered will be the first frame acquired after the start of the space link session.

3.4.2.6 stop-time

3.4.2.6.1 The value of the **stop-time** parameter shall be ‘null’, or it shall be a time value that indicates that data transfer should cease when the next frame that would be transferred has an ERT later than **stop-time**.

3.4.2.6.2 For the online delivery mode, only frames acquired during the space link session associated with this service instance shall be delivered, regardless of **stop-time**.

3.4.2.6.3 For the offline delivery mode, the provider shall deliver all available frames that meet the delivery criteria regardless of the space link session in which they were acquired.

3.4.2.6.4 For the online delivery mode, if **stop-time** is ‘null’, the provider shall continue to transfer all frames that are acquired from the space link and satisfy the delivery criteria until either the user invokes an RCF-STOP operation or the association is released or aborted.

3.4.2.6.5 For the offline delivery mode, **stop-time** must not be ‘null’.

3.4.2.6.6 To be valid, **stop-time** must satisfy the following criteria:

- a) **stop-time** must be later than the **start-time** (see 3.4.2.5);
- b) for the online delivery mode, **stop-time**, if not ‘null’, must be earlier than or equal to the end time of the service instance provision period for this service instance;
- c) for the offline delivery mode, **stop-time** plus the offline processing latency must be earlier than the current time.

NOTES

- 1 Offline processing latency is the length of time after a frame is acquired from the space link before it is available for retrieval using the offline delivery mode. The actual value of offline processing latency is negotiated between SLE Complex Management and SLE Utilization Management.

- 2 Offline delivery is only available for frames which already have been acquired when the RCF-START operation is invoked.

3.4.2.7 global-VCID

3.4.2.7.1 The **global-VCID** parameter shall identify the master channel or virtual channel that is to be provided to the user and shall consist of the frame version number, the spacecraft identifier, and the virtual channel identifier.

NOTES

- 1 The definitions of spacecraft identifier and virtual channel identifier depend on the frame version number. If the frame version number indicates that the virtual channel consists of transfer frames, then the definitions of spacecraft identifier and virtual channel identifier are as per reference [3]. If the frame version number indicates that the virtual channel consists of VCDUs or CVCDUs, then the definitions of spacecraft identifier and virtual channel identifier are as per reference [4].
- 2 The physical channel is not specified directly through the RCF service. Rather, the selection of physical channel is determined through the service package, which specifies the RAF service instance that is consumed by the RFP-FG that is producing the RCF service.

3.4.2.7.2 The frame version number shall be a valid frame version number defined by CCSDS.

NOTE – At the time of issuance of this Recommendation, the only valid frame version numbers were ‘00’ (version 1) and ‘01’ (version 2) (see references [3] and [4]).

3.4.2.7.3 The spacecraft identifier shall be a valid spacecraft identifier as defined by CCSDS (see references [3] and [4]).

3.4.2.7.4 The virtual channel identifier shall be a valid virtual identifier as defined by CCSDS (see references [3] and [4]) or it shall be the value ‘any’. The value ‘any’ indicates that a master channel, defined by the frame version number and the spacecraft identifier, shall be provided by the RCF service. Otherwise, a virtual channel shall be provided by the RCF service.

3.4.2.8 result

The **result** parameter shall specify the result of the RCF-START operation and shall contain one of the following values:

- a) ‘positive result’—the RCF-START operation has been performed by the provider, and the provider shall henceforth invoke RCF-TRANSFER-DATA operations as needed to transfer to the user all available frames that meet the specified delivery criteria;

- b) 'negative result'—the RCF-START operation has not been performed by the provider, and the provider shall not invoke any RCF-TRANSFER-DATA operations even if frames are available.

3.4.2.9 diagnostic

3.4.2.9.1 If `result` is 'negative result', `diagnostic` shall be present and shall contain one of the following values:

- a) 'duplicate Invoke-ID'—the value of the `invoke-ID` parameter is the same as the `invoke-ID` of a previous, outstanding operation;
- b) 'out of service'—the provider has been taken out of service for an indefinite period by management action;
- c) 'unable to comply'—the provider is unable to transfer data at this time because of a fault affecting the provider;
- d) 'invalid start time'—the value of the `start-time` provided in the invocation is not valid;
- e) 'invalid stop time'—the value of the `stop-time` provided in the invocation is not valid;
- f) 'missing time value'—for the offline delivery mode, the value of `start-time` and/or `stop-time` was 'null';
- g) 'invalid global-VCID'—the value specified for the `global-VCID` parameter is not valid;
- h) 'other'—the reason for the negative result will have to be found by other means.

3.4.2.9.2 If `result` is 'positive result', `diagnostic` shall not be present.

3.4.3 EFFECTS

3.4.3.1 If `result` is 'positive result':

- a) the provider shall transition to state 3 ('active');
- b) in the 'active' state, the provider shall transfer frames to the user whenever they are available and satisfy the delivery criteria.

3.4.3.2 If `result` is 'negative result':

- a) the provider shall remain in state 2 ('ready') and shall not deliver frames even if they are available;
- b) if the diagnostic is 'unable to comply',

- 1) the user may re-invoke the RCF-START operation at a later time within the constraints of the service instance provision period;
- 2) if the provider's SLE complex management determines that being 'unable to comply' is more than a transient problem, it shall cause the provider to invoke the RCF-PEER-ABORT operation.

3.5 RCF-STOP

3.5.1 PURPOSE

3.5.1.1 The user shall invoke the RCF-STOP operation to request that the provider stop delivering telemetry frames.

NOTE – Within the constraints of the service provision period, the user may re-enable frame delivery by invoking the RCF-START operation.

3.5.1.2 The provider shall provide a report of the outcome of the performance of the RCF-STOP operation to the user.

3.5.1.3 RCF-STOP is valid only in state 3 ('active') and shall be invoked only by the user.

3.5.2 INVOCATION, RETURN, AND PARAMETERS

3.5.2.1 General

The parameters of the RCF-STOP operation shall be present in the invocation and return as specified in table 3-4.

Table 3-4: RCF-STOP Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
result		M
diagnostic		C

3.5.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the RCF-STOP invocation (see 3.1.5).

3.5.2.3 performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of RCF-STOP (see 3.1.5).

3.5.2.4 **invoke-ID**

The RCF service provider shall return unchanged the user-supplied value of the **invoke-ID** parameter (see 3.1.6).

3.5.2.5 **result**

The **result** parameter shall specify the result of the RCF-STOP operation and shall contain one of the following values:

- a) 'positive result'—the RCF-STOP operation has been performed by the provider, and the delivery of telemetry frames to the user has ceased;
- b) 'negative result'—the RCF-STOP operation has not been performed by the provider for the reason specified by the **diagnostic** parameter, and the delivery of telemetry frames to the user continues.

3.5.2.6 **diagnostic**

3.5.2.6.1 If **result** is 'negative result', **diagnostic** shall be present and its value shall be one of the following:

- a) 'duplicate Invoke-ID'—the value of the **invoke-ID** parameter is the same as the **invoke-ID** of a previous, outstanding operation;
- b) 'other'—the reason for the negative result will have to be found by other means.

3.5.2.6.2 If **result** is 'positive result', **diagnostic** shall not be present.

3.5.3 **EFFECTS**

3.5.3.1 If **result** is 'positive result', the provider shall:

- a) cease invoking RCF-TRANSFER-DATA operations;
- b) discard the contents of transfer buffer;
- c) transition to state 2 ('ready').

3.5.3.2 If **result** is 'negative result', the provider remains in state 3, and processing continues unchanged.

3.6 RCF-TRANSFER-DATA

3.6.1 PURPOSE

3.6.1.1 The provider shall invoke the RCF-TRANSFER-DATA operation to deliver a telemetry frame to the user.

3.6.1.2 The RCF-TRANSFER-DATA operation shall be an unconfirmed operation.

NOTE – Although RCF-TRANSFER-DATA is an unconfirmed operation, it is assumed that the communications service provides certain guarantees, as described in 1.3.1.

3.6.1.3 RCF-TRANSFER-DATA is valid only in state 3 ('active') and shall be invoked only by the provider.

3.6.2 INVOCATION, RETURN, AND PARAMETERS

3.6.2.1 General

The parameters of the RCF-TRANSFER-DATA operation shall be present in the invocation as specified in table 3-5.

Table 3-5: RCF-TRANSFER-DATA Parameters

Parameters	Invocation
invoker-credentials	M
earth-receive-time	M
antenna-ID	M
data-link-continuity	M
private-annotation	M
data	M

3.6.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the user to authenticate the RCF-TRANSFER-DATA invocation (see 3.1.5).

3.6.2.3 earth-receive-time

The **earth-receive-time** parameter shall contain the UTC time at which the signal event corresponding to the leading edge of the first bit of the telemetry frame was presented at the phase center of the antenna used to acquire the frame.

NOTE – The first bit of the frame is the first bit following the attached sync marker.

3.6.2.4 **antenna-ID**

3.6.2.4.1 The **antenna-ID** parameter shall indicate which antenna of the SLE Complex was used to acquire this frame of data.

NOTE – **antenna-ID** is provided specifically to identify the physical location used as the reference point for the **earth-receive-time** parameter.

3.6.2.4.4 SLE Complex Management and SLE Utilization Management shall mutually agree upon the allowable values for **antenna-ID** and their interpretation.

NOTE – It is assumed that **antenna-ID** is an identifier that may be used to look up the actual location information, which is provided outside the scope of this service.

3.6.2.5 **data-link-continuity**

3.6.2.5.1 The **data-link-continuity** parameter shall indicate whether this frame was the direct successor on the space link of the previous frame acquired by the instance of RAF production associated with this service instance.

NOTE – Even though this frame may not have been the direct successor of the previous frame acquired from the space link, that does not necessarily imply that any frames from this master channel or virtual channel are missing (e.g., because the missing frames belonged to a different master channel or virtual channel).

3.6.2.5.2 The **data-link-continuity** parameter shall contain an integer value:

- a) A value of ‘-1’ shall indicate that this is the first frame after the start of production;
- b) a value of ‘0’ shall indicate that this frame is the direct successor to the last frame acquired from the space link by RAF production;
- c) any non-zero positive value shall indicate that this frame is not the direct successor to the last frame acquired from the space link:
 - 1) a non-zero positive value further indicates an estimate of the number of frames that were missed since the last frame acquired before this frame;
 - 2) a value of ‘1’ may be used if no better estimate is available.

3.6.2.6 private-annotation

The **private-annotation** parameter shall be used to convey additional information that may be associated with a frame:

- a) it may be set to 'null' to indicate that there is no private annotation;
- b) if not 'null', there must be a prior arrangement between SLE Complex Management and SLE Utilization Management regarding the contents and interpretation of this parameter.

3.6.2.7 data

The **data** parameter shall consist of a transfer frame or a VCDU from the master channel or virtual channel that was requested by the user.

NOTES

- 1 The attached sync marker is not delivered as part of **data**.
- 2 For a Reed-Solomon encoded frame (i.e., a CVCDU), the Reed-Solomon check symbols are discarded, and the frame less the check symbols (i.e., a VCDU) is delivered to the user.

3.6.3 EFFECTS

The effects of the RCF-TRANSFER-DATA operation shall be as follows:

- a) A telemetry frame acquired by the provider from the requested master channel or virtual channel shall be delivered to the user.
- b) The provider shall remain in state 3 ('active').

3.7 RCF-SYNC-NOTIFY

3.7.1 PURPOSE

3.7.1.1 The RCF service provider shall invoke the RCF-SYNC-NOTIFY operation to notify the user of the occurrence of an event affecting the production of the RCF service.

NOTE – Notification of events may be of value to the user in understanding specific provider behavior, such as an interruption in frame delivery.

3.7.1.2 The RCF-SYNC-NOTIFY operation shall be an unconfirmed operation.

3.7.1.3 The order in which the RCF-SYNC-NOTIFY and RCF-TRANSFER-DATA operations are invoked shall reflect the actual chronology of events.

NOTE – For example, if an RCF-SYNC-NOTIFY operation is invoked after one RCF-TRANSFER-DATA operation but before another, then the event indicated by the notification occurred after the ERT of the frame associated with the preceding RCF-TRANSFER-DATA but before the ERT of the frame associated with the following RCF-TRANSFER-DATA.

3.7.1.4 RCF-SYNC-NOTIFY is valid only in state 3 ('active') and shall be invoked only by the provider.

3.7.2 INVOCATION, RETURN, AND PARAMETERS

3.7.2.1 General

The parameters of the RCF-SYNC-NOTIFY operation shall be present in the invocation as specified in table 3-6.

Table 3-6: RCF-SYNC-NOTIFY Parameters

Parameter	Invocation
invoker-credentials	M
notification-type	M
notification-value	C

3.7.2.2 **invoker-credentials**

The **invoker-credentials** parameter shall provide information that enables the user to authenticate the RCF-SYNC-NOTIFY invocation (see 3.1.5).

3.7.2.3 **notification-type**

The **notification-type** parameter shall describe the event being notified to the user, and its value shall be one of the following:

- a) 'loss of frame synchronization'—the frame synchronization process is not able to synchronize to the stream of frames from the space link:
 - 1) the notification shall be invoked at most once for each occurrence of a frame synchronizer transition from 'in-lock' to 'out-of-lock';
 - 2) the provider shall minimize the latency from the time the loss of frame synchronization event occurs until the notification is invoked;
 - 3) there shall be no explicit notification when the frame synchronizer transitions from 'out-of-lock' back to 'in-lock'; rather, the next invocation of RCF-TRANSFER-DATA shall implicitly convey the occurrence of that event;
 - 4) loss of frame synchronization notifications shall not be invoked in the offline delivery mode;

NOTE – Because this notification refers to processing of frames from the space link, it may or may not indicate that frames were lost on the master channel or virtual channel being provided by this instance of service.

- b) 'production status change'—the status of RCF production has changed from 'running' to 'halted' or vice versa:
 - 1) a change in production status shall be the result of action taken by service management;
 - 2) production status change notifications shall not be invoked in the offline delivery mode;
- c) 'data discarded due to excessive backlog'—some data was discarded by the RCF service provider, either because of timeliness considerations (timely online mode) or because of online frame buffer overflow (complete online mode):
 - 1) if the contents of the online frame buffer are discarded two or more times in a row without a successful intervening delivery of frames to the user, no more than one data discarded notification shall be delivered to the user;

- 2) data discarded notifications shall not be invoked in the offline delivery mode;
- d) 'end of data'—the provider has no more data to send.

NOTE – The 'end of data' notification is invoked in all delivery modes. For example, for an online service instance, the space link session has ended, and there are no more frames to be delivered; or, for an offline service instance, all available frames between the specified start and end times (see 3.4) have been delivered).

3.7.2.4 notification-value

3.7.2.4.1 The presence of the **notification-value** parameter in the return from RCF-SYNC-NOTIFY shall be conditional on the value of **notification-type**.

3.7.2.4.2 If **notification-type** is 'loss of frame synchronization', then **notification-value** shall be present and shall convey the following information:

- a) the UTC time when the frame synchronizer transitioned from 'in-lock' to 'out-of-lock';
- b) the current status of the carrier demodulation process, which shall be 'in-lock', 'out-of-lock', 'not in use', or 'unknown';
- c) the current status of the subcarrier demodulation process, which shall be 'in-lock', 'out-of-lock', 'not in use', or 'unknown';
- d) the current status of the symbol synchronization process, which shall be 'in-lock', 'out-of-lock', 'not in use', or 'unknown'.

NOTE – The determinations of the lock statuses of carrier demodulation, subcarrier demodulation, and symbol synchronization typically are based on measurements that are integrated over some time period. To that extent, the values reported here may reflect the statuses of the corresponding processes at a time slightly earlier than the time when the notification is invoked.

3.7.2.4.3 If **notification-type** is 'production status change', then the **notification-value** parameter shall be present, and its value shall indicate the current production status, which shall be one of the following:

- a) 'running'—the RCF production process is capable of processing a return space link physical channel, if available;
- b) 'halted'—the RCF production process is inactive and unable to process a return space link physical channel until action is taken by service management.

3.7.2.4.4 If **notification-type** is 'data discarded due to excessive backlog' or 'end of data', the **notification-value** parameter shall not be present.

3.7.3 EFFECTS

The effects of the RCF-SYNC-NOTIFY operation shall be as follows:

- a) Information about the occurrence of the specified event shall be delivered to the user.
- b) The state of the provider shall not change.

3.8 RCF-SCHEDULE-STATUS-REPORT

3.8.1 PURPOSE

3.8.1.1 The user shall invoke the RCF-SCHEDULE-STATUS-REPORT operation to request that the provider send a status report either immediately or periodically or to stop the sending of such reports.

3.8.1.2 The provider shall return a report of the outcome of the performance of the RCF-SCHEDULE-STATUS-REPORT operation to the user.

3.8.1.3 The provider shall send the requested status report(s) by means of the RCF-STATUS-REPORT operation (see 3.9).

3.8.1.4 For periodic reporting, the user may change the reporting period by invoking another RCF-SCHEDULE-STATUS-REPORT.

3.8.1.5 RCF-SCHEDULE-STATUS-REPORT is valid only in states 2 ('ready') and 3 ('active') and shall be invoked only by the user.

3.8.2 INVOCATION, RETURN, AND PARAMETERS

3.8.2.1 General

The parameters of the RCF-SCHEDULE-STATUS-REPORT operation shall be present in the invocation and return as specified in table 3-7.

Table 3-7: RCF-SCHEDULE-STATUS-REPORT Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
report-request-type	M	
reporting-cycle	C	
result		M
diagnostic		C

3.8.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the RCF-SCHEDULE-STATUS-REPORT invocation (see 3.1.5).

3.8.2.3 performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of RCF-SCHEDULE-STATUS-REPORT (see 3.1.5).

3.8.2.4 invoke-ID

The performer shall return unchanged the invoker-supplied value of the **invoke-ID** parameter (see 3.1.6).

3.8.2.5 report-request-type

3.8.2.5.1 The **report-request-type** parameter shall specify how reporting shall be done, and its value shall be one of the following:

- a) 'immediately'—send a single status report immediately;
- b) 'periodically'—send a status report every **reporting-cycle** seconds;
- c) 'stop'—do not send further status reports.

3.8.2.5.2 If **report-request-type** is 'immediately',

- a) the provider shall stop sending status reports after the immediate status report has been sent;
- b) periodic reporting may be restarted by means of another RCF-SCHEDULE-STATUS-REPORT operations.

3.8.2.6 reporting-cycle

3.8.2.6.1 If the value of the **report-request-type** parameter is 'periodically', then **reporting-cycle** shall be present and shall specify the requested interval between status reports in seconds.

3.8.2.6.2 If the value of the **report-request-type** parameter is not 'periodically', the **reporting-cycle** parameter shall not be present.

3.8.2.7 result

The **result** parameter shall specify the result of the RCF-SCHEDULE-STATUS-REPORT operation, and its value shall be one of the following:

- a) 'positive result'—the RCF-SCHEDULE-STATUS-REPORT operation has been performed, and the provider will send the requested report(s) or stop sending periodic status reports;
- b) 'negative result'— the RCF-SCHEDULE-STATUS-REPORT operation has not been performed for the reason specified in the `diagnostic` parameter. The previous setting for status reporting remains in effect.

3.8.2.8 diagnostic

If `result` is 'negative result', `diagnostic` shall be provided and shall contain one of the following values:

- a) 'duplicate Invoke-ID'—the value of the `invoke-ID` parameter is the same as the `invoke-ID` of a previous, outstanding operation;
- b) 'already stopped'—the provider is not currently providing periodic reports (when `report-request-type` is 'stop');
- c) 'invalid reporting cycle'—the requested `reporting-cycle` value is outside the range agreed upon by management arrangement;
- d) 'other'—the reason for rejection of the operation will have to be found by other means.

3.8.2.8.3 If `result` is 'positive result', `diagnostic` shall not be present.

3.8.3 EFFECTS

3.8.3.1 If `result` is 'positive result':

- a) if `reporting-request-type` specifies 'immediately':
 - 1) a status report shall be sent immediately;
 - 2) the sending of any previously requested periodic status reports shall cease;
- b) if `reporting-request-type` is 'periodically':
 - 1) a status report shall sent immediately;
 - 2) subsequent status reports shall be sent at the interval specified in the `reporting-cycle` parameter;
- c) if `reporting-request-type` is 'stop', periodic status reporting shall cease.

3.8.3.2 If `result` is 'negative result', the RCF-SCHEDULE-STATUS-REPORT shall have no effect, and the previous setting for status reporting shall remain in effect.

3.8.3.3 The state of the provider shall remain unchanged.

3.9 RCF-STATUS-REPORT

3.9.1 PURPOSE

3.9.1.1 The provider shall invoke the RCF-STATUS-REPORT operation to send a status report to the user.

3.9.1.2 RCF-STATUS-REPORT shall be an unconfirmed operation.

3.9.1.3 Status reports shall be sent (or not sent) in accordance with user requests conveyed by means of the RCF-SCHEDULE-STATUS-REPORT operation (see 3.8).

3.9.1.4 The RCF-STATUS-REPORT operation is valid only in states 2 ('ready') and 3 ('active') and shall be invoked only by the provider.

3.9.2 INVOCATION, RETURN, AND PARAMETERS

3.9.2.1 General

The parameters of the RCF-STATUS-REPORT operation shall be present in the invocation as specified in table 3-8.

Table 3-8: RCF-STATUS-REPORT Parameters

Parameters	Invocation
invoker-credentials	M
number-of-frames-delivered	M
frame-sync-lock-status	M
symbol-sync-lock-status	M
subcarrier-lock-status	M
carrier-lock-status	M
production-status	M

3.9.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the RCF-STATUS-REPORT invocation (see 3.1.5).

3.9.2.3 number-of-frames-delivered

The **number-of-frames-delivered** parameter shall specify the total number of telemetry frames that have been delivered to the user since the start of the service instance provision period.

3.9.2.4 frame-sync-lock-status

The **frame-sync-lock-status** parameter shall specify the current lock status of the frame synchronization process, the value of which shall be 'in-lock', 'out-of-lock', 'not in use', or 'unknown'.

3.9.2.5 symbol-sync-lock-status

The **symbol-sync-lock-status** parameter shall specify the current lock status of the symbol (or bit) synchronization process, the value of which shall be 'in-lock', 'out-of-lock', 'not in use', or 'unknown'.

3.9.2.6 subcarrier-lock-status

The **subcarrier-lock-status** parameter shall specify the current lock status of the subcarrier demodulation process, the value of which shall be 'in-lock', 'out-of-lock', 'not in use', or 'unknown'.

3.9.2.7 carrier-lock-status

The **carrier-lock-status** parameter shall specify the current lock status of the carrier demodulation process, the value of which shall be 'in-lock', 'out-of-lock', 'not in use', or 'unknown'.

3.9.2.8 production-status

The **production-status** parameter shall specify the current status of RCF production, the value of which shall be 'running' or 'halted'.

NOTE – See 3.7 for descriptions of the **production-status** values.

3.9.3 EFFECTS

The effects of the RCF-STATUS-REPORT operation shall be as follows:

- a) Status information shall be delivered to the user.
- b) The state of the provider shall not change.

3.10 RCF-GET-PARAMETER

3.10.1 PURPOSE

3.10.1.1 The user shall invoke the RCF-GET-PARAMETER operation to ascertain the value of an RCF service parameter.

3.10.1.2 The provider shall return a report of the outcome of the performance of the RCF-GET-PARAMETER operation to the user.

3.10.1.3 If the operation is successful, the current value of the specified RCF service parameter shall be provided to the user in the return from the operation.

3.10.1.4 RCF-GET-PARAMETER is valid in state 2 ('ready') and state 3 ('active') and shall be invoked only by the user.

3.10.2 INVOCATION, RETURN, AND PARAMETERS

3.10.2.1 General

The parameters of the RCF-GET-PARAMETER operation shall be present in the invocation and return as specified in table 3-9.

Table 3-9: RCF-GET-PARAMETER Parameters

Parameters	Invocation	Return
invoker-credentials	M	
performer-credentials		M
invoke-ID	M	M
rcf-parameter	M	C
parameter-value		C
result		M
diagnostic		C

3.10.2.2 invoker-credentials

The **invoker-credentials** parameter shall provide information that enables the performer to authenticate the RCF-GET-PARAMETER invocation (see 3.1.5).

3.10.2.3 performer-credentials

The **performer-credentials** parameter shall provide information that enables the invoker to authenticate the return from the performance of RCF-GET-PARAMETER (see 3.1.5).

3.10.2.4 invoke-ID

The performer shall return unchanged the invoker-supplied value of the **invoke-ID** parameter (see 3.1.6).

3.10.2.5 rcf-parameter

3.10.2.5.1 The **rcf-parameter** parameter shall specify the RCF service parameter whose value is to be returned to the user and shall contain one of the following values:

- a) 'latency limit'—the maximum allowable delivery latency time in seconds for the timely online delivery mode (i.e., the maximum delay between reception of a frame on the ground and its delivery to the user); the value of this parameter is undefined if the delivery mode is not timely online;
- b) 'transfer buffer size'—the size of the transfer buffer in bytes.
- c) 'delivery mode'—the delivery mode for this instance of RCF service: it is set by service management, and the permitted values are 'timely online delivery mode', 'complete online delivery mode', and 'offline delivery mode';
- d) 'reporting cycle'—'null' if cyclic reporting is off, otherwise the time (in seconds) between successive RCF-STATUS-REPORT invocations (see 3.8);
- e) 'return timeout period'—the maximum time period (in seconds) permitted from invocation of a confirmed RCF operation until the return must be received by the invoker (see 4.1.3).

3.10.2.5.2 The **rcf-parameter** parameter is conditionally present in the return based on the **result** parameter:

- a) if the value of **result** is 'positive result', **rcf-parameter** shall be present;
- b) if the value of **result** is 'negative result', **rcf-parameter** shall not be present.

3.10.2.6 parameter-value

3.10.2.6.1 The **parameter-value** parameter shall contain the value for the parameter indicated by **rcf-parameter** as described in 3.10.2.5.

3.10.2.6.2 The `parameter-value` parameter is conditionally present in the return based on the `result` parameter:

- a) if the value of `result` is 'positive result', `parameter-value` shall be present;
- b) if the value of `result` is 'negative result', `parameter-value` shall not be present.

3.10.2.7 result

The **result** parameter shall specify the result of the RCF-GET-PARAMETER operation and shall contain one of the following values:

- a) 'positive result'—the RCF-GET-PARAMETER operation has been performed, and the value of the specified RCF service parameter is provided in the return to the user;
- b) 'negative result'—the RCF-GET-PARAMETER operation has not been performed for the reason specified in the `diagnostic` parameter.

3.10.2.8 diagnostic

3.10.2.8.1 If `result` is 'negative result', the `diagnostic` parameter shall be present, and its value shall be one of the following:

- a) 'duplicate Invoke-ID'—the value of the `invoke-ID` parameter is the same as the `invoke-ID` of a previous, outstanding operation;
- b) 'unknown parameter'—the parameter specified by the value of `rcf-parameter` is not known to the service provider;
- c) 'other'—the reason for the negative result will have to be found by other means.

3.10.2.8.2 If `result` is 'positive result', `diagnostic` shall not be present.

3.10.3 EFFECTS

3.10.3.1 If `result` is 'positive result', the value of the parameter specified by the value of `rcf-parameter` shall be returned to the user in the return.

3.10.3.2 If `result` is 'negative result', the value of the parameter specified by the value of `rcf-parameter` shall not be returned to the user.

3.10.3.3 The state of the provider shall not change.

3.11 RCF-PEER-ABORT

3.11.1 PURPOSE

3.11.1.1 The user or provider shall invoke the RCF-PEER-ABORT operations to notify the peer system that the local application detected an error that requires that the association between them to be terminated.

3.11.1.2 RCF-STATUS-REPORT shall be an unconfirmed operation.

3.11.1.3 RCF-PEER-ABORT is valid only in states 2 ('ready') and 3 ('active') and may be invoked by either the user or the provider.

3.11.2 INVOCATION, RETURN, AND PARAMETERS

3.11.2.1 General

The parameters of the RCF-PEER-ABORT operation shall be present in the invocation as specified in table 3-10.

Table 3-10: RCF-PEER-ABORT Parameters

Parameters	Invocation
diagnostic	M

3.11.2.2 diagnostic

The **diagnostic** parameter shall specify why the RCF-PEER-ABORT is being invoked, and its value shall be one of the following:

- a) 'access denied'—a responder with an identity as presented in the **responder-identifier** parameter of the RCF-BIND return is not known to the initiator (e.g., the value of the **responder-identifier** parameter does not match the authorized responder for any service instance known to the initiator).
- b) 'unexpected responder ID'—the value of the **responder-identifier** parameter in the RCF-BIND return does not match the identity of the authorized responder for this service instance as specified by service management.
- c) 'operational requirement'—the local system had to terminate the association to accommodate some other operational need;
- d) 'protocol error'—the local application detected an error in the sequencing of RCF service operations;
- e) 'communications failure'—the communications service on the other side of a gateway was disrupted;

NOTE – This **diagnostic** value is only applicable when the SLE applications are communicating via a gateway.

- f) ‘encoding error’—the local application detected an error in the encoding of one or more operation parameters or did not recognize the operation;
- g) ‘return timeout’—the local application detected that the return from a confirmed operation was not received within a specified period of time;
- h) ‘end of service provision period’—the local application detected that the service instance provision period has ended and the initiator has not invoked the RCF-UNBIND operation;
- i) ‘unsolicited invoke-ID’—the local application received a return with an invoke-ID that does not match the invoke-ID of any of the operations for which a return is pending;
- j) ‘other’—the local application detected an unspecified error during the processing of one or more operations.

NOTE – RCF-PEER-ABOT does not carry an **invoker-credentials** parameter. It is conceivable that an intruder may use the RCF-PER-ABORT operation for a denial-of-service attack. If an intruder has that capability, then a denial-of-service attack can be much more easily accomplished by disrupting communications at a layer lower than the applications layer. Therefore, authentication of RCF-PEER-ABORT would not provide improved protections against such attacks.

3.11.3 EFFECTS

The effects of the RCF-PEER-ABORT operation shall be as follows:

- a) The association shall be aborted, and the user and the provider shall cease to communicate with each other.

NOTE – The RCF-PEER-ABORT operation shall be truly abortive. This implies that the transfer of the SLE-PDU carrying this invocation shall not be delayed by other invocations submitted earlier.

- b) The provider shall transition to state 1 (‘unbound’).
- c) The provider shall discard the contents of the transfer buffer.
- d) The values of RCF service parameters shall return to the initial values set by service management, including the values of any parameters changed by the service user.
- e) Statistical information required for the generation of the status report shall be retained throughout the service instance provision period.

4 RCF PROTOCOL

4.1 GENERIC PROTOCOL CHARACTERISTICS

4.1.1 UNEXPECTED PROTOCOL DATA UNIT

If the peer application invokes an operation not allowed in the current state of the performer, the performer shall abort the association by invoking the RCF-PEER-ABORT operation with the `diagnostic` parameter set to 'protocol error'.

4.1.2 INVALID PROTOCOL DATA UNIT

If the application receives an invocation or return that contains an unrecognized operation type, contains a parameter of the wrong type, or is otherwise not decodable, the application shall abort the association by invoking the RCF-PEER-ABORT operation with the `diagnostic` parameter set to 'encoding error'.

4.1.3 MISSING RETURN

For confirmed operations:

- a) If the invoker does not receive the return from the performer within a timeout period specified by service management, the invoker shall abort the association by invoking the RCF-PEER-ABORT operation with the `diagnostic` parameter set to 'return timeout'.
- b) The timeout period shall be chosen taking into account performance of user and provider applications as well as the delays introduced by the underlying communications service.

NOTE – In order to provide responsive service and short timeout periods, the generation of the return from an operation must not depend on any human interaction.

4.1.4 UNSOLICITED RETURN

If the application receives a return with an `invoke-ID` parameter value that does not correspond to any invocation for which a return is still pending, the application shall abort the association by invoking the RCF-PEER-ABORT operation with the `diagnostic` parameter set to 'unsolicited Invoke-ID'.

4.1.5 COMMUNICATIONS FAILURE

4.1.5.1 If an SLE entity, i.e., an SLE user or provider, is informed by its underlying communications service that the connection with the peer entity has been closed due to a communications failure, then this entity shall consider that the association with the peer entity has been aborted.

4.1.5.2 Occurrence of the above described communications failure event shall be referred to as a 'protocol abort'.

NOTE – The exact criteria for determining when communications have failed and the communications connection is closed depend on the characteristics of the communications technology used and may be specific to a given implementation. However, it is a requirement on any communications service underlying an SLE transfer service that it include suitable means to ascertain the status of the connection (see 1.3.1). In the case of communications protocols that do not intrinsically provide such a capability, the transmission of a periodic 'heartbeat' indicator or equivalent may need to be implemented.

4.1.5.3 The SLE entity shall be able to accept additional diagnostic information that the communications service provides together with the notification of the 'protocol abort' event.

NOTE – Such diagnostic information is assumed to be descriptive of the specific nature or cause of the fault. Its type and format will be implementation dependent. The SLE entity may log this information, display it to the user, etc.

4.1.5.4 Subsequent to a 'protocol abort' event:

- a) the RCF provider shall transition to state 1 ('unbound');
- b) neither the user nor the provider shall attempt further communications with the peer except that the initiator may attempt to re-establish the association by invoking the RCF-BIND operation;
- c) the provider shall discard the contents of the transfer buffer;
- d) the values of RCF service parameters shall return to the initial values set by service management, including the values of any parameters changed by the service user; and
- e) statistical information required for the generation of the status report shall be retained throughout the service instance provision period.

4.1.6 ACCESS CONTROL

4.1.6.1 The initiator of an association shall present its own identity in the `initiator-identifier` parameter of the RCF-BIND invocation.

4.1.6.2 If the RCF-BIND operation is invoked with a value of `initiator-identifier` that is not known to the responder, the responder shall not make any attempt to authenticate that invocation. Instead, the responder shall generate an RCF-BIND return with `result` set to 'negative result', `diagnostic` set to 'access denied', and `performer-credentials` set to 'unused'.

4.1.6.3 If the value of `initiator-identifier` is known to the responder, the responder shall attempt to authenticate the RCF-BIND invocation (see 3.1.5) as required for

the given initiator. If authentication succeeds but the initiator is not the authorized initiator for the service instance indicated in the `service-instance-identifier` parameter of the RCF-BIND invocation, the responder shall generate an RCF-BIND return with `result` set to 'negative result' and `diagnostic` set to 'service instance not accessible to this initiator'.

NOTE – If authentication fails, the responder shall behave as specified in 4.1.6.2.

4.1.6.4 If the initiator receives an RCF-BIND return with a `responder-identifier` value that is not known to the initiator, the initiator shall not make any attempt to authenticate this return but shall abort the association by invoking RCF-PEER-ABORT with `diagnostic` set to 'access denied'.

4.1.6.5 If the initiator receives an RCF-BIND return with a `responder-identifier` value that is known to the initiator, the initiator shall attempt to authenticate the RCF-BIND return (see 3.1.5) as required for the given responder. If authentication succeeds but the `responder-identifier` is not the authorized initiator for this service instance as specified by service management, the initiator shall abort the association by means of the RCF-PEER-ABORT operation with `diagnostic` set to 'unexpected responder ID'.

NOTE – If authentication fails, the initiator shall behave as specified in 4.1.6.4.

4.1.7 FAILING AUTHENTICATION

4.1.7.1 An incoming invocation or return shall be ignored if the `credentials` parameter cannot be authenticated when, by management arrangement, credentials are required.

4.1.7.2 If an invocation is ignored, the operation shall not be performed, and a report of the outcome shall not be returned to the invoker.

4.1.7.3 If a return is ignored, it shall be as if no report of the outcome of the operation has been received.

4.2 STATE TRANSITION TABLE

4.2.1 The behavior of the RCF service provider shall conform to the state transition matrix specified in table 4-1.

4.2.2 The state transition matrix specified in table 4-1 represents one instance of service and thus one association. Once the association is established, if an RCF-BIND invocation for a different association but for the same service instance is received, it shall be rejected with an RCF-BIND return with the `result` parameter set to 'negative result' and the `diagnostic` parameter set to 'already bound'. This event shall not affect the association already in place.

NOTES

- 1 The state table specifies operation interactions and state transitions for the service provider in its role as either initiator or responder.
- 2 The leftmost column of the state table lists all incoming events. Where these events correspond to the arrival of an incoming SLE-PDU, the ASN.1 type defined for this SLE-PDU in Annex A is indicated in parentheses ().
- 3 Where an event is internal to the provider, its description is put in single quotation marks ‘ ’. These events are defined in table 4-2.
- 4 The three columns (one column per state) on the right side of the table specify the behavior the provider will exhibit, which depends on the current state and the incoming event. In some cases, the behavior in addition depends on Boolean conditions, also referred to as predicates. Such conditions are put in double quotation marks “ ”. The predicates are defined in table 4-3. Predicates that are simple Boolean variables set only by that state machine itself are referred to as Boolean flags and specified in table 4-4. The dependency on a predicate is presented in form of an IF <condition> THEN <action> [ELSEIF <condition> THEN <action>] ELSE <action> clause.
- 5 If the action given in the table is simply to send a specific SLE-PDU, that is indicated by the appearance of the name of ASN.1 type of the SLE-PDU to be sent in parentheses (). If that SLE-PDU is a return, the name may be preceded by the plus symbol (+) to indicate that result is ‘positive result’ or by the negative symbol (-) to indicate ‘negative result’. Where several actions are to be taken (referred to as a ‘compound action’), the name of the compound action is put in curly braces { }. The individual actions making up each compound action are identified in table 4-5.
- 6 ‘Not applicable’ is stated where the given event can only occur in the given state because of an implementation error on the provider side.
- 7 Where the consequences of an incoming event are not visible to the user because the provider does not send any SLE-PDU in reaction to the given event, the action is put in square brackets [].
- 8 State transitions are indicated by an arrow and the number of the state that will be entered; for example, → 1 indicates the transition to state 1.
- 9 The actions to be taken and the state transition are considered to be one atomic action. The sequence shown in the table is irrelevant except that SLE-PDUs shall be sent in the sequence stated in the table.
- 10 Whenever the provider invokes a confirmed operation with invoke-ID set to <n>, it shall start an associated return <n> timer. Should this timer expire before the return <n> is received, the provider shall invoke RCF-PEER-ABORT.

Table 4-1: Provider Behavior

No.	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
1	'start of service instance provision period'	IF "provider initiated" THEN {invoke bind} → 1 ELSE [ignore] → 1	Not applicable	Not applicable
2	'return <n> timer expired'	IF "bind pending" THEN {return time-out} → 1 IF NOT 'end of service instance provision period' THEN {invoke bind} ELSE [ignore] ELSE Not applicable → 1	{peer abort 'return time-out'} → 1	{peer abort 'return time-out'} → 1
3	(-rcfBindReturn)	IF "bind pending" THEN set "bind pending" FALSE → 1 stop return <n> timer IF "retry permitted" THEN {invoke bind} ELSE release resources ELSE [ignore] → 1	{peer abort 'protocol error'} → 1	{peer abort 'protocol error'} → 1
4	(+rcfBindReturn)	IF "bind pending" THEN set "bind pending" FALSE → 2 stop return <n> timer IF NOT "compatible" THEN {invoke unbind} ELSE [ignore] ELSE [ignore] → 1	{peer abort 'protocol error'} → 1	{peer abort 'protocol error'} → 1
5	(rcfBindInvocation)	IF "provider initiated" THEN [ignore] → 1 ELSE IF "positive result" THEN (+rcfBindReturn) → 2 ELSE (-rcfBindReturn) → 1	{peer abort 'protocol error'} → 1	{peer abort 'protocol error'} → 1

Table 4-1: Provider Behavior (continued)

No.	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
6	'end of service instance provision period'	[ignore]	IF "provider initiated" THEN {invoke unbind} → 2 ELSE {peer abort 'end of service instance provision period'} → 1	{peer abort 'end of service instance provision period'} → 1
7	(rcfUnbindReturn)	[ignore]	IF "unbind pending" THEN {provider unbind} → 1 IF "done" THEN release resources ELSE [ignore] ELSE {peer abort 'protocol error'} → 1	{peer abort 'protocol error'} → 1
8	(rcfUnbindInvocation)	[ignore]	IF "provider initiated" THEN {peer abort 'protocol error'} → 1 ELSE {user unbind} → 1 IF "end" THEN release resources ELSE [ignore]	{peer abort 'protocol error'} → 1
9	(rcfStartInvocation)	[ignore]	IF "unbind pending" THEN {peer abort 'protocol error'} → 1 ELSE IF "positive result" THEN (+rcfStartReturn) → 3 initialize transfer buffer ELSE (-rcfStartReturn) → 2	{peer abort 'protocol error'} → 1
10	(rcfStopInvocation)	[ignore]	{peer abort 'protocol error'} → 1	IF "positive result" THEN stop release timer → 2 (+rcfStopReturn) ELSE (-rcfStopReturn) → 3
11	'data available', "complete online" delivery mode or "offline" delivery mode	Not applicable	Not applicable	IF "buffer full" THEN submit buffer contents UNTIL (NOT "congested") → 3 reinitialize transfer buffer {insert annotated frame} ELSE {insert annotated frame} → 3

Table 4-1: Provider Behavior (continued)

No.	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
12	'data available', "timely online" delivery mode	Not applicable	Not applicable	IF "buffer full" THEN {pass buffer contents} → 3 IF "congested" THEN {sync notify 'data discarded'} {start release timer} {insert annotated frame} ELSE {insert annotated frame} {start release timer} ELSE IF "buffer empty" THEN {insert annotated frame} → 3 {start release timer} ELSE {insert annotated frame} → 3
13	'release timer expired', "timely online" delivery mode	Not applicable	Not applicable	{pass buffer contents} → 3 IF "congested" THEN {sync notify 'data discarded'} {start release timer} ELSE [ignore]
14	'end of data', "timely online" delivery mode	Not applicable	Not applicable	IF "buffer full" THEN {pass buffer contents} → 3 IF "congested" THEN {sync notify 'data discarded'} {sync notify 'end of data'} submit buffer contents UNTIL (NOT "congested") ELSE {sync notify 'end of data'} submit buffer contents UNTIL (NOT "congested") ELSE IF "buffer empty" THEN {sync notify 'end of data'} → 3 submit buffer contents UNTIL (NOT "congested") ELSE stop release timer → 3 {sync notify 'end of data'} submit buffer contents UNTIL (NOT "congested")

Table 4-1: Provider Behavior (continued)

No.	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
15	'end of data', "complete online" delivery mode or "offline" delivery mode	Not applicable	Not applicable	IF "buffer full" THEN submit buffer contents UNTIL (NOT "congested") → 3 reinitialize transfer buffer {sync notify 'end of data'} submit buffer contents UNTIL (NOT "congested") ELSE {sync notify 'end of data'} → 3 submit buffer contents UNTIL (NOT "congested")
16	'loss of frame synchronization', "timely online" delivery mode	Not applicable	[ignore] → 2	IF "buffer full" THEN {pass buffer contents} → 3 IF "congested" THEN {sync notify 'data discarded'} {start release timer} {sync notify 'loss of frame sync'} ELSE {sync notify 'loss of frame sync'} {start release timer} ELSE IF "buffer empty" THEN {sync notify 'loss of frame sync'} → 3 {start release timer} ELSE (sync notify 'loss of frame sync') → 3
17	'loss of frame synchronization', "complete online" delivery mode	Not applicable	[ignore] → 2	IF "buffer full" THEN submit buffer contents UNTIL (NOT "congested") → 3 reinitialize transfer buffer {sync notify 'loss of frame sync'} ELSE {sync notify 'loss of frame sync'} → 3
18	'production status change', "timely online" delivery mode or "complete online" delivery mode	Not applicable	IF NOT "unbind pending") THEN {sync notify 'production status change'} → 2 ELSE [ignore] → 2	{sync notify production status change} → 3

Table 4-1: Provider Behavior (continued)

No.	Incoming Event	Unbound (State 1)	Ready (State 2)	Active (State 3)
19	(rcfScheduleStatusReportInvocation)	[ignore]	IF "positive result" THEN (+rcfScheduleStatusReport) → 2 IF "immediately" THEN {immediate report} ELSE IF "periodically" THEN {periodic report} ELSE stop reporting-cycle timer ELSE → 2 (-rcfScheduleStatusReportReturn)	IF "positive result" THEN (+rcfScheduleStatusReport) → 3 IF "immediately" THEN {immediate report} ELSE IF "periodically" THEN {periodic report} ELSE stop reporting-cycle timer ELSE → 3 (-rcfScheduleStatusReportReturn)
20	'reporting-cycle timer expired'	Not applicable	{periodic report} → 2	{periodic report} → 3
21	(rcfGetParameterInvocation)	[ignore]	IF "positive result" THEN (+rcfGetParameterReturn) → 2 ELSE (-rcfGetParameterReturn) → 2	IF "positive result" THEN (+rcfGetParameterReturn) → 3 ELSE (-rcfGetParameterReturn) → 3
22	(rcfPeerAbortInvocation)	[ignore]	{clean up} → 1	{clean up} → 1
23	'invalid SLE-PDU'	[ignore]	{peer abort ('encoding error')} → 1	{peer abort ('encoding error')} → 1
24	'return SLE-PDU with unsolicited Invoke-ID'	[ignore]	{peer abort ('unsolicited Invoke-ID')} → 1	{peer abort ('unsolicited Invoke-ID')} → 1
25	'protocol abort'	[ignore]	{clean up} → 1	{clean up} → 1
26	'not authenticated SLE-PDU'	[ignore] → 1	[ignore] → 2	[ignore] → 3

Table 4-2: Event Description References

Event	Reference
'data available'	3.4.1.3
'end of data'	3.7.2.3
'end of service instance provision period'	3.11.2.2
'invalid SLE-PDU'	4.1.2
'loss of frame synchronization'	3.7.2.3
'not authenticated SLE-PDU'	4.1.7
'production status change'	3.7.2.3
'release timer expired'	3.1.8.1.1
'reporting-cycle timer expired'	3.8.2.6
'return SLE-PDU with unsolicited Invoke-ID'	4.1.4
'return <n> timer expired'	4.1.3
'start of service instance provision period'	3.2.1.2

Table 4-3: Predicate Descriptions

Predicate	Evaluates to TRUE if
"buffer empty"	There are no RCF SLE-PDUs in the transfer buffer
"buffer full"	The transfer buffer cannot accommodate the currently available annotated frame or synchronous notification
"compatible"	The version number and the application context contained in (+rcfBindReturn) are supported by the provider
"complete online"	Delivery mode is complete online
"congested"	The underlying communications service cannot accept the contents of the transfer buffer due to congestion
"done"	The unbind-reason parameter value in the provider-initiated BIND invocation was 'end'
"end"	All checks on the UNBIND invocation are passed and the unbind-reason parameter value is 'end'
"immediately"	All parameter checks on the RCF-SCHEDULE-STATUS-REPORT are passed and the report-request-type value is 'immediately'
"offline"	Delivery mode is offline
"online"	Delivery mode is timely online or complete online
"periodically"	All parameter checks on the RCF-SCHEDULE-STATUS-REPORT are passed and the report-request-type value is 'periodically'
"positive result"	All checks on the invocation are passed
"provider initiated"	The RCF-BIND operation is specified to be initiated by the provider for this service instance

Predicate	Evaluates to TRUE if
"retry permitted"	The diagnostic value contained in the (-rcfBindReturn) is 'unable to comply' or 'other', and the service instance provision period is still active
"timely online"	Delivery mode is timely online

Table 4-4: Boolean Flags

Flag Name	Initial Value
"bind pending"	FALSE
"unbind pending"	FALSE

Table 4-5: Compound Actions Definitions

Name	Actions Performed
{clean up}	stop release timer stop all return timers stop reporting-cycle timer reinitialize transfer buffer reset parameter values to those specified in service package
{immediate report}	(rcfStatusReportInvocation) stop reporting-cycle timer
{insert annotated frame}	annotate the available frame with the parameters of the RCF-TRANSFER-DATA operation insert the annotated frame into the transfer buffer
{invoke bind}	(rcfBindInvocation) set "bind pending" to TRUE start return <n> timer
{invoke unbind}	(rcfUnbindInvocation) stop reporting-cycle timer set "unbind pending" to TRUE start return <n> timer
{pass buffer contents}	stop release timer submit contents of transfer buffer to underlying communications service reinitialize transfer buffer

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Name	Actions Performed
{peer abort 'xxxx'}	stop release timer stop all return timers stop reporting-cycle timer reinitialize transfer buffer (rcfPeerAbortInvocation) with diagnostic set to 'xxxx'
{periodic report}	(rcfStatusReportInvocation) set reporting-cycle timer to the reporting-cycle value in the most recent SCHEDULE-STATUS-REPORT invocation start reporting-cycle timer
{provider unbind}	set "unbind pending" to FALSE stop all return timers
{return time-out}	(rcfPeerAbortInvocation) with diagnostic 'return time-out' set "bind pending" to FALSE set "unbind pending" to FALSE
{start release timer}	set release timer to latency limit start release timer
{sync notify 'xxxx'}	create an RCF synchronous notification with diagnostic set to 'xxxx' insert the notification into the transfer buffer
{user unbind}	stop reporting-cycle timer stop all return timers (rcfUnbindReturn)

ANNEX A

DATA TYPE DEFINITIONS

(This annex is part of the Recommendation)

A1 INTRODUCTION

This annex defines the data types that are used by the RCF service. It is intended to provide a clear specification of these data types and to avoid ambiguity. It is not intended to constrain how these data types are implemented or encoded. These definitions are suitable for inclusion in any type of ASN.1 based protocol that implements the RCF service.

The data type definitions are presented in six ASN.1 modules.

Subsection A 2.1 contains basic types that are common with other SLE Transfer Services. As more services become specified by CCSDS, further types may be added to this module or existing types may be extended. However, that eventuality is not expected to invalidate the module in its present form because it is expected that an implementation compliant with a future extended version of this module will be interoperable with an implementation based on its present version.

Subsection A 2.2 specifies the SLE-PDUs exchanged between an SLE user and an SLE provider application in order to establish, release or abort an association. They are common among SLE transfer service types.

Subsection A 2.3 specifies SLE-PDUs related to invocations and returns that are common to SLE transfer service types.

Subsection A 2.4 specifies data types specific to the RCF service. In part, these specific types are derived from types specified in A 2.1 by means of subtyping.

Subsection A 2.5 specifies all incoming (from a provider point of view) SLE-PDUs. Where applicable, these SLE-PDUs are mapped to the generic SLE-PDUs defined in A 2.2 and A 2.3.

Subsection A 2.6 specifies in the same way the outgoing SLE-PDUs.

Although subsections A 2.2, A 2.3, A 2.5 and A 2.6 define the SLE-PDUs that will be exchanged between the SLE provider and user applications, they shall not be interpreted as requiring that these SLE-PDUs shall be completely mapped to the user data field of the underlying communications protocol. For example, depending on the communications

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protocol(s) used, part of the SLE-PDUs may be used to determine the appropriate setting of protocol control information.

A2 RCF DATA TYPE SPECIFICATION**A2.1 SLE TRANSFER SERVICE COMMON TYPES**

CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

EXPORTS      ConditionalTime
,            Credentials
,            DeliveryMode
,            Diagnostics
,            Duration
,            ForwardDuStatus
,            IntPosLong
,            IntPosShort
,            IntUnsignedLong
,            IntUnsignedShort
,            InvokeId
,            ParameterName
,            ProductionStatus
,            Report
,            SpaceLinkDataUnit
,            Time
;

```

```

ConditionalTime ::= CHOICE
{
    undefined    [0] NULL
,
    known        [1] Time
}

```

```

-- If credentials are used, it will be necessary that
-- the internal structure of the octet string is known
-- to both parties. Since the structure will depend on
-- algorithm used, it is not specified here. However,
-- the peer entities may use ASN.1 encoding to make the
-- internal structure visible.

```

```

Credentials ::= CHOICE
{
    unused      [0] NULL
,
    used        [1] OCTET STRING (SIZE (8 .. 256))
}

```

```

DeliveryMode ::= INTEGER
{
    rtnTimelyOnline      (0)
,
    rtnCompleteOnline    (1)
,
    rtnOffline           (2)
,
    fwdOnline            (3)
,
    fwdOffline           (4)
}

```

```

Diagnostics ::= INTEGER
{
    duplicateInvokeId    (100)
,
    otherReason          (101)
}

```

```

-- The Duration is expressed in microseconds
Duration ::= IntUnsignedLong

```

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```

ForwardDuStatus      ::=  INTEGER
{
    radiated          (0)
    , expired         (1)
    , interrupted     (2)
    , transferred     (3)    -- FSP
    , productionStarted (4)
    , undefined       (5)    -- FSP
}

-- 1 to (2^32)-1
IntPosLong           ::=  INTEGER (1 .. 42949667295)

-- 1 to (2^16)-1
IntPosShort          ::=  INTEGER (1 .. 65535)

-- 0 to (2^32)-1
IntUnsignedLong      ::=  INTEGER (0 .. 42949667295)

-- 0 to (2^16)-1
IntUnsignedShort     ::=  INTEGER (0 .. 65535)

InvokeId             ::=  IntUnsignedShort

ParameterName        ::=  INTEGER
{
    aggregationTimeoutPeriod (0)
    , aggregationUsage       (1)
    , apidList                (2)
    , bitLock                 (3)
    , bitLockRequired         (4)
    , bitRate                 (5)
    , bufferSize              (6)
    , copInEffect              (7)
    , deliveryMode             (8)
    , directiveInvocation      (9)
    , expectedDirectiveIdentification (10)
    , expectedSlduIdentification (11)
    , fecfUsage                (12)
    , fopSlidingWindow        (13)
    , fopState                 (14)
    , idlePattern              (15)
    , latencyLimit             (16)
    , mapList                  (17)
    , mapMuxControl            (18)
    , mapMuxScheme             (19)
    , maximumFrameLength       (20)
    , maximumPacketLength      (21)
    , maximumSlduLength        (22)
    , modulationIndex          (23)
    , plopInEffect             (24)
    , reportingCycle           (25)
    , requestedFrameQuality     (26)
    , returnTimeoutPeriod      (27)
    , rfAvailable              (28)
    , rfAvailableRequired      (29)
    , segmentHeader            (30)
    , sentQueueSize            (31)
    , subcarrierFrequency      (32)
    , timeoutType              (33)
    , timerInitial             (34)
    , transmissionLimit        (35)
    , vcMuxControl             (36)
    , vcMuxScheme              (37)
    , virtualChannel           (38)
}

```

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```
ProductionStatus      ::=  INTEGER
{
    running            (0)
  ,
    waiting            (1)
  ,
    interrupted        (2)
  ,
    halted             (3)
}

Report                ::=  INTEGER
{
    produceReport      (0)
  ,
    doNotProduceReport (1)
}

SpaceLinkDataUnit     ::=  OCTET STRING (SIZE (7 .. 65536))

Time                  ::=  CHOICE
{
    ccsdsFormat        [0]  TimeCCSDS
}

TimeCCSDS              ::=  OCTET STRING (SIZE(8))
-- P-field is implicit (not present, defaulted to 41 hex
-- T-field:
-- 2 octets: number of days since 1958/01/01 00:00:00
-- 4 octets: number of milliseconds of the day
-- 2 octets: number of microseconds of the millisecond
--           (set to 0 if not used)
-- This definition reflects exactly the format of the CCSDS defined
-- time tag as used in spacelink data units.

END
```

A2.2 SLE TRANSFER SERVICE BIND TYPES

CCSDS-SLE-TRANSFER-SERVICE-BIND-TYPES

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

EXPORTS      SleBindInvocation
,            SleBindReturn
,            SlePeerAbort
,            SleUnbindInvocation
,            SleUnbindReturn
;

```

```

IMPORTS      Credentials
,            IntPosShort
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES
;

```

```

-- =====
-- The first part of the module definition contains the SLE-PDUs
-- =====

```

```

SleBindInvocation      ::= SEQUENCE
{
    invokerCredentials      Credentials
,    initiatorIdentifier    AuthorityIdentifier
,    responderPortIdentifier PortId
,    serviceType            ApplicationIdentifier
,    versionNumber          VersionNumber
,    serviceInstanceIdentifier ServiceInstanceIdentifier
}

```

```

SleBindReturn          ::= SEQUENCE
{
    performerCredentials    Credentials
,    responderIdentifier    AuthorityIdentifier
,    result                 CHOICE
    {
        positive            [0] VersionNumber
        negative            [1] BindDiagnostic
    }
}

```

```

SlePeerAbort           ::= PeerAbortDiagnostic

```

```

SleUnbindInvocation    ::= SEQUENCE
{
    invokerCredentials      Credentials
,    unbindReason           UnbindReason
}

```

```

SleUnbindReturn        ::= SEQUENCE
{
    responderCredentials    Credentials
,    result                 CHOICE
    {
        positive            [0] NULL
    }
}

```

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```

-- =====
-- The second part of the module definition contains the types
-- used by the SLE-PDUs declared in the first part.
-- =====

ApplicationIdentifier ::= INTEGER
{
    rtnAllFrames          (0)
  , rtnInsert            (1)
  , rtnChFrames          (2)
    -- rtnChFrames includes rtnMcFrames and rtnVcFrames
  , rtnChFsh             (3)
    -- rtnChFsh includes rtnMcFsh and rtnVcFsh
  , rtnChOcf             (4)
    -- rtnChOcf includes rtnMcOcf and rtnVcOcf
  , rtnBitstr            (8)  -- AOS
  , fwdBitstr            (9)
  , fwdProtoVcdu         (10)
  , fwdInsert            (11)
  , fwdCVcdu             (12)
  , fwdTcSpacePkt        (13) -- conventional telecommand
  , fwdTcVca             (14) -- conventional telecommand
  , fwdTcFrame           (15)
  , fwdCltu              (16)
  , rtnAllFramesLtd      (20)
  , fwdCltuLtd           (21)
}

ATTRIBUTE ::= CLASS
{
    &attributeId          OBJECT IDENTIFIER UNIQUE
  , &AttributeType
}
-- The actual constraints are specified in reference [E4].
-- &AttributeType is of the type VisibleString
-- except for the attribute specifying the delivery
-- mode. The latter is of the type INTEGER.

AttributeTypeAndValue ::= SEQUENCE
{
    attributeID           ATTRIBUTE.&attributeId
  , attributeValue        ATTRIBUTE.&AttributeType
}

AuthorityIdentifier ::= VisibleString (SIZE (6 .. 16))

BindDiagnostic ::= INTEGER
{
    accessDenied          (0)
  , serviceTypeNotSupported (1)
  , versionNotSupported   (2)
  , noSuchServiceInstance (3)
  , alreadyBound          (4)
  , siNotAccessibleToThisInitiator (5)
  , inconsistentServiceType (6)
  , invalidTime           (7)
  , unableToComply        (8)
  , outOfService          (9)
  , otherReason           (1000)
}

LogicalPortName ::= VisibleString (SIZE (1 .. 128))

```

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```

PeerAbortDiagnostic      ::=  INTEGER
{
    accessDenied          (0)
  , unexpectedResponderId (1)
  , operationalRequirement (2)
  , protocolError        (3)
  , communicationsFailure (4)
  , encodingError         (5)
  , returnTimeout         (6)
  , endOfServiceProvisionPeriod (7)
  , unsolicitedInvokeId   (8)
  , other                 (127)
-- The range of this INTEGER shall be (0 .. 255).
-- (128 .. 255) is reserved for diagnostic codes
-- that are specific to the communications technology used.
}

PortId                  ::=  LogicalPortName

RDNSequence             ::=  SEQUENCE OF RelativeDistinguishedName

RelativeDistinguishedName ::=  SET SIZE (1) OF AttributeTypeAndValue

ServiceInstanceIdentifier ::=  RDNSequence

UnbindReason            ::=  INTEGER
{
    end                  (0)
  , suspend              (1)
  , versionNotSupported (2)
  , other                (1000)
}

VersionNumber           ::=  IntPosShort

END

```

A2.3 SLE TRANSFER COMMON PDUs

CCSDS-SLE-TRANSFER-SERVICE-COMMON-PDUS

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

EXPORTS          ReportingCycle
,                SleAcknowledgement
,                SleScheduleStatusReportInvocation
,                SleScheduleStatusReportReturn
,                SleStopInvocation
;

```

```

IMPORTS          Credentials
,                Diagnostics
,                InvokeId
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES
;

```

```

-- =====
-- The first part of the module definition contains the SLE-PDUs
-- =====

```

```

SleAcknowledgement ::= SEQUENCE
{
    credentials      Credentials
,   invokeId        InvokeId
,   result          CHOICE
    {
        positiveResult [0] NULL
        ,
        negativeResult [1] Diagnostics
    }
}

SleScheduleStatusReportInvocation ::= SEQUENCE
{
    invokerCredentials Credentials
,   invokeId          InvokeId
,   reportRequestType ReportRequestType
}

SleScheduleStatusReportReturn ::= SEQUENCE
{
    performerCredentials Credentials
,   invokeId          InvokeId
,   result          CHOICE
    {
        positiveResult [0] NULL
        ,
        negativeResult [1] DiagnosticScheduleStatusReport
    }
}

SleStopInvocation ::= SEQUENCE
{
    invokerCredentials Credentials
,   invokeId          InvokeId
}

```

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```
-- =====
-- The second part of the module definition contains the types
-- used by the SLE-PDUs declared in the first part.
-- =====
```

```
DiagnosticScheduleStatusReport ::= CHOICE
{
    common          [0]  Diagnostics
  ,
    specific
    {
        alreadyStopped          (0)
        invalidReportingCycle    (1)
    }
}
```

```
-- The cycle duration is expressed in seconds
ReportingCycle ::= INTEGER (60 .. 600)
```

```
ReportRequestType ::= CHOICE
{
    immediately      [0]  NULL
  ,
    periodically     [1]  ReportingCycle
  ,
    stop              [2]  NULL
}
```

```
END
```

A2.4 RCF TRANSFER STRUCTURES

CCSDS-SLE-TRANSFER-SERVICE-RCF-STRUCTURES

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

EXPORTS      AntennaId
,            DiagnosticRcfGet
,            DiagnosticRcfStart
,            GvcId
,            LockStatus
,            Notification
,            RcfGetParameter
,            RcfProductionStatus
,            RequestedFrameQuality
;
IMPORTS      Diagnostics
,            IntPosShort
,            IntPosLong
,            ParameterName
,            ProductionStatus
,            Time
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES
;

AntennaId ::= CHOICE
{
    globalForm      [0] OBJECT IDENTIFIER
,
    localForm       [1] OCTET STRING (SIZE (1 .. 16))
}

DiagnosticRcfGet ::= CHOICE
{
    common          [0] Diagnostics
,
    specific        [1] INTEGER
    {
        unknownParameter (0)
    }
}

DiagnosticRcfStart ::= CHOICE
{
    common          [0] Diagnostics
,
    specific        [1] INTEGER
    {
        outOfService (0)
        ,
        unableToComply (1)
        ,
        invalidStartTime (2)
        ,
        invalidStopTime (3)
        ,
        missingTimeValue (4)
    }
}

GvcId ::= SEQUENCE
{
    spacecraftId    INTEGER (0 .. 1023)
,
    versionNumber   INTEGER (0 .. 3)
,
    vcId            INTEGER (0 .. 63)
}
-- Note that the permissible range of the vcId depends on the

```

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```
--versionNumber.
```

```

LockStatus                                ::=  INTEGER
{
    inLock                                (0)
,    outOfLock                            (1)
,    notInUse                             (2)
,    unknown                             (3)
}

LockStatusReport                          ::=  SEQUENCE
{
    time                                  Time
,    carrierLockStatus                   LockStatus
,    subcarrierLockStatus                LockStatus
,    symbolSyncLockStatus                 LockStatus
}

Notification                              ::=  CHOICE
{
    lossFrameSync                        [0]  LockStatusReport
,    productionStatusChange              [1]  RcfProductionStatus
,    excessiveDataBacklog                 [2]  NULL
,    endOfData                           [3]  NULL
}

RcfGetParameter                           ::=  CHOICE
(
    parLatencyLimit                      [1]  SEQUENCE
    {
        parameterName                    ParameterName (latencyLimit)
,        parameterValue                   CHOICE
        {
            timelyOnline                  [0]  IntPosShort
,            notTimelyOnline               [1]  NULL
        }
    }
,    parBufferSize                        [2]  SEQUENCE
    {
        parameterName                    ParameterName (bufferSize)
,        parameterValue                   CHOICE
        {
            timelyOnline                  [0]  IntPosLong
,            notTimelyOnline               [1]  NULL
        }
    }
)

RcfProductionStatus                       ::=  ProductionStatus
(
    running
|    halted
)

END

```

A2.5 RCF INCOMING PDUs

CCSDS-SLE-TRANSFER-SERVICE-RCF-INCOMING-PDUS

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

IMPORTS      Credentials
,            InvokeId
,            ConditionalTime
,            ParameterName
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES

            SleScheduleStatusReportInvocation
,            SleStopInvocation
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-PDUS

            GVcId
FROM CCSDS-SLE-TRANSFER-SERVICE-RCF-STRUCTURES

            SleBindInvocation
,            SleBindReturn
,            SlePeerAbort
,            SleUnbindInvocation
,            SleUnbindReturn
FROM CCSDS-SLE-TRANSFER-SERVICE-BIND-TYPES
;

```

```

-- =====
-- The first part of the module definition contains the RCF type
-- that contains all the possible PDUs the provider may receive.
-- =====

```

```

RcfUserToProviderPdu      ::= CHOICE
{      rcfBindInvocation      [100] SleBindInvocation
,      rcfBindReturn          [101] SleBindReturn
,      rcfUnbindInvocation     [102] SleUnbindInvocation
,      rcfUnbindReturn        [103] SleUnbindReturn
,      rcfStartInvocation      [0]   RcfStartInvocation
,      rcfStopInvocation       [1]   SleStopInvocation
,      rcfScheduleStatusReportInvocation [2]
                                      SleScheduleStatusReportInvocation
,      rcfGetParameterInvocation [3]   RcfGetParameterInvocation

```

```

,      rcfPeerAbortInvocation      [104] SlePeerAbort
}

-- =====
-- The second part of the module definition contains the types
-- used by the RCF-PDUs declared in the first part.
-- =====
RcfGetParameterInvocation      ::= SEQUENCE
{
    invokerCredentials          Credentials
,
    invokeId                    InvokeId
,
    rcfParameter                ParameterName
    (
        latencyLimit
    |
        bufferSize
    )
}

RcfStartInvocation             ::= SEQUENCE
{
    invokerCredentials          Credentials
,
    invokeId                    InvokeId
,
    startTime                    ConditionalTime
,
    stopTime                    ConditionalTime
,
    gVcId                       GVcId
}

END

```

A2.6 RCF OUTGOING PDUS

CCSDS-SLE-TRANSFER-SERVICE-RCF-OUTGOING-PDUS

DEFINITIONS

IMPLICIT TAGS

::= BEGIN

```

IMPORTS      Credentials
,            IntUnsignedLong
,            InvokeId
,            SpaceLinkDataUnit
,            Time
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-TYPES

            SleAcknowledgement
,            SleScheduleStatusReportReturn
FROM CCSDS-SLE-TRANSFER-SERVICE-COMMON-PDUS

            AntennaId
,            DiagnosticRcfGet
,            DiagnosticRcfStart
,            LockStatus
,            Notification
,            RcfGetParameter
,            RcfProductionStatus
FROM CCSDS-SLE-TRANSFER-SERVICE-RCF-STRUCTURES

            SleBindInvocation
,            SleBindReturn
,            SlePeerAbort
,            SleUnbindInvocation
,            SleUnbindReturn
FROM CCSDS-SLE-TRANSFER-SERVICE-BIND-TYPES
;
-- =====
-- The first part of the module definition contains the RCF type
-- that contains all the possible PDUs the provider may send.
-- =====

RcfProviderToUserPdu ::= CHOICE
{
    rcfBindInvocation      [100] SleBindInvocation
,
    rcfBindReturn          [101] SleBindReturn
,
    rcfUnbindInvocation    [102] SleUnbindInvocation
,
    rcfUnbindReturn        [103] SleUnbindReturn
,
    rcfStartReturn         [0]   RcfStartReturn
,
    rcfStopReturn          [1]   SleAcknowledgement

```

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```

,      rcfTransferBuffer          [2]    RcfTransferBuffer
,      rcfScheduleStatusReportReturn [3]    SleScheduleStatusReportReturn
,      rcfStatusReportInvocation    [4]    RcfStatusReportInvocation
,      rcfGetParameterReturn        [5]    RcfGetParameterReturn
,      rcfPeerAbortInvocation        [104] SlePeerAbort
}

```

```

-- =====
-- The second part of the module definition contains the types
-- used by the RCF-PDUs declared in the first part.
-- =====

```

```

FrameOrNotification ::= CHOICE
{
    annotatedFrame      [0]    RcfTransferDataInvocation
,
    syncNotification    [1]    RcfSyncNotifyInvocation
}

```

```

RcfGetParameterReturn ::= SEQUENCE
{
    performerCredentials Credentials
,
    invokeId             InvokeId
,
    result               CHOICE
    {
        positiveResult [0]    RcfGetParameter
    ,
        negativeResult [1]    DiagnosticRcfGet
    }
}

```

```

RcfStartReturn ::= SEQUENCE
{
    performerCredentials Credentials
,
    invokeId             InvokeId
,
    result               CHOICE
    {
        positiveResult [0]    NULL
    ,
        negativeResult [1]    DiagnosticRcfStart
    }
}

```

```

RcfStatusReportInvocation ::= SEQUENCE
{
    invokerCredentials Credentials
,
    deliveredFrameNumber IntUnsignedLong
,
    frameSyncLockStatus  LockStatus
,
    symbolSyncLockStatus LockStatus
,
    subcarrierLockStatus LockStatus
,
    carrierLockStatus    LockStatus
,
    productionStatus     RcfProductionStatus
}

```

```

RcfTransferBuffer ::= SEQUENCE OF FrameOrNotification

```

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```

RcfSyncNotifyInvocation ::= SEQUENCE
{
    invokerCredentials    Credentials
,
    notification          Notification
}

RcfTransferDataInvocation ::= SEQUENCE
{
    invokerCredentials    Credentials
,
    earthReceiveTime      Time
,
    antennaId             AntennaId
,
    dataLinkContinuity    INTEGER (-1 .. 65535)
,
    privateAnnotation     CHOICE
        {
            null          [2] NULL
            ,
            notNull       [1] OCTET STRING (SIZE ( 1 ..128))
        }
,
    data                  SpaceLinkDataUnit
}

END

```

ANNEX B**INDEX TO DEFINITIONS**

(This annex **is not** part of the Recommendation)

This annex lists terms used in this Recommendation and, for each term, provides a reference to the definition of that term.

Term	Reference
(data) type	reference [7]
(data) value	reference [7]
abstract binding	reference [1]
abstract object	reference [1]
abstract port	reference [1]
abstract service	reference [1]
abstract syntax	reference [6]
Abstract Syntax Notation One (ASN.1)	reference [7]
active (state)	subsection 2.6.4.2
application entity	reference [6]
application layer	reference [6]
application process	reference [6]
association	subsection 1.6.1.9.1
attached sync marker	reference [2]
authentication algorithm	subsection 3.1.5
authentication level	subsection 3.1.5
authorized initiator	subsection 3.1.4
authorized responder	subsection 3.1.4
blocking operation	subsection 3.1.6.1
codeblock	reference [2]
coded virtual channel data unit (CVCDU)	reference [4]
communications service	subsection 1.6.1.9.2

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complete (online delivery mode)	subsections 2.3, 3.1.8.2
concatenation	reference [6]
confirmed operation	subsection 1.6.1.9.3
convolutional code	reference [2]
credentials	subsection 3.1.5
cyclic redundancy check (CRC)	reference [4]
distinguished name (DN)	reference [9]
error control field	subsection 1.6.1.9.4
flow control	reference [6]
frame	subsection 1.6.1.9.14
frame error control field	reference [3]
frame version number	subsection 1.6.1.9.5
initiator	reference [8]
invocation	subsection 1.6.1.9.6
invoker	reference [1]
latency limit	subsections 2.6.4.6.1, 3.1.8.1
master channel	subsection 1.6.1.9.7
Mission Data Operation System (MDOS)	reference [1]
Mission User Entity (MUE)	reference [1]
non-blocking operation	subsection 3.1.6.1
object identifier	reference [7]
offline delivery mode	reference [1]
offline frame buffer	subsections 2.6.4.6.2, 3.1.8.3
online delivery mode	reference [1]
online frame buffer	subsections 2.6.4.6.2, 3.1.8.2
Open System Interconnection (OSI)	reference [6]
operation	reference [1]
parameter	subsection 0
performance	subsection 1.6.1.9.10
performer	reference [1]
physical channel	reference [1]
port identifier	subsection 1.6.1.9.10
provider-initiated	subsection 2.3

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pseudo-randomization	reference [2]
ready (state)	subsection 2.6.4.2
real system	reference [6]
Reed-Solomon check symbols	reference [2]
Reed-Solomon code	reference [2]
relative distinguished name (RDN)	reference [9]
release timer	subsections 2.6.4.6.1, 3.1.8.1
responder	reference [8]
return (of an operation)	subsection 1.6.1.9.11
Return Channel Frames channel	reference [1]
Return Channel Frames service	reference [1]
return data	reference [1]
service access point (SAP)	reference [6]
service instance provision period	subsection 1.6.1.9.12
service provider (provider)	reference [1]
service user (user)	reference [1]
SLE Complex	reference [1]
SLE Complex Management	reference [1]
SLE data channel	reference [1]
SLE functional group (SLE-FG)	reference [1]
SLE protocol data unit (SLE-PDU)	reference [1]
SLE service data unit (SLE-SDU)	reference [1]
SLE service package	reference [1]
SLE System	reference [1]
SLE transfer service instance	reference [1]
SLE transfer service production	reference [1]
SLE transfer service provision	reference [1]
SLE Utilization Management	reference [1]
spacecraft identifier	1.6.1.9.13
space link	reference [1]
space link data channel	reference [1]
space link data unit (SL-DU)	reference [1]
space link session	reference [1]

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telemetry frame	subsection 1.6.1.9.14
timely (online delivery mode)	subsections 2.3, 3.1.8.1
transfer buffer	subsections 2.6.4.6.1, 3.1.8
transfer frame	reference [3]
transfer syntax	reference [6]
unbound (state)	subsection 2.6.4.2
unconfirmed operation	subsection 1.6.1.9.15
user-initiated	subsection 2.3
VCDU error control field	reference [4]
virtual channel	subsection 1.6.1.9.16
virtual channel data unit (VCDU)	reference [4]
virtual channel identifier (VCID)	subsection 1.6.1.9.17

ANNEX C**ACRONYMS**

(This annex **is not** part of the Recommendation)

This annex lists the acronyms used in this Recommendation.

ASDC	Abstract Service Definition Conventions
ASN.1	Abstract Syntax Notation One
C	Conditional
CCSDS	Consultative Committee for Space Data Systems
CM-P	Complex Management Port
ERT	Earth Receive Time
FG	Functional Group
FSH	Frame Secondary Header
IP	Internet Protocol
ISO	International Organization for Standardization
M	Mandatory
MDOS	Mission Data Operation System
MUE	Mission User Entity
OCF	Operations Control Field
PDU	Protocol Data Unit
RAF	Return All Frames
RCF	Return Channel Frames
RF	Radio Frequency
RFP-FG	Return Frame Processing Functional Group

RSLP-FG	Return Space Link Processing Functional Group
SAP	Service Access Point
SDU	Service Data Unit
SLE	Space Link Extension
SLE-FG	SLE Functional Group
SLE-PDU	SLE Protocol Data Unit
SLE-SDU	SLE Service Data Unit
SL-DU	Space Link Data Unit
TCP	Transmission Control Protocol
T-P	Transfer Provider Port
T-U	Transfer User Port
UTC	Coordinated Universal Time

ANNEX D**CONFORMANCE MATRIX**

(This annex **is not** part of the Recommendation)

This annex provides the Conformance Matrix for implementations of RCF service. An RCF service will be considered to be 'conformant' with this Recommendation if the mandatory elements identified in the Tables D-1 and D-2 are implemented as described in this Recommendation.

Table D-1: Conformance Matrix for RCF Service (Operations)

RCF Service Operation	Optional/Mandatory
RCF-BIND RCF-UNBIND	These operations are mandatory. All parameters are also mandatory with the exception of: Invoker-credentials Performer-credentials
RCF-START RCF-STOP RCF-TRANSFER-DATA RCF-SYNC-NOTIFY RCF-SCHEDULE-STATUS-REPORT RCF-STATUS-REPORT RCF-GET-PARAMETER	These operations are mandatory. All parameters are also mandatory with the exception of: Invoker-credentials Performer-credentials
RCF-PEER-ABORT	This operation is mandatory. All parameters are also mandatory with the exception of: Invoker-credentials

Table D-2: Conformance Matrix for RCF Service (Other Requirements)

Requirement	Optional / Mandatory
Online buffer size (3.1.8)	Mandatory minimum size is 1024 maximum-sized telemetry frames
State transition table (4.2)	Mandatory
ASN.1 protocol specification (annex A)	All data types are mandatory. Implementations may vary as described in the annex.

ANNEX E

INFORMATIVE REFERENCES

(This annex **is not** part of the Recommendation)

- [E1] *Procedures Manual for the Consultative Committee for Space Data Systems*. CCSDS A00.0-Y-7. Yellow Book. Issue 7. Washington, D.C.: CCSDS, November 1996.
- [E2] *Standard Terminology, Conventions, and Methodology (TCM) for Defining Data Services*. Report Concerning Space Data Systems Standards, CCSDS 910.2-G-1. Green Book. Issue 1. Washington, D.C.: CCSDS, November 1994.
- [E3] *Space Link Extension Service Management Specification*. Draft Recommendation for Space Data System Standards, CCSDS 910.5-W-1.18. White Book. Issue 1.20. CCSDS, July 1999.
- [E4] *Cross Support Concept – Part 1: Space Link Extension Services*. Report Concerning Space Data Systems Standards. CCSDS 910.3-G-1. Green Book. Issue 1. Washington, D.C.: CCSDS, May 1995.
- [E5] *Space Link Extension—Return All Frames Service Specification*. Draft Recommendation for Space Data System Standards, CCSDS 911.1-R-2. Red Book. Issue 2. Washington, D.C.: CCSDS, November 1999.